

Ireland's electricity network from public data

What was built, where every number comes from, and what the result supports

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Abstract

This repository builds two things from published data. The first is a measurement of how much of Ireland's electricity network OpenStreetMap actually contains, which decides whether volunteer mapping can stand in for the distribution operator's own records. The second is a working power-system model: EirGrid's four published planning cases converted into PyPSA networks, given coordinates from OpenStreetMap, cut down to a study region, fitted with a week of hourly profiles, and packaged so that somebody else can run them.

This document is the guide to both. Its first job is provenance. Every figure quoted here traces to a named file, produced by a named function, from a source that is named with its publisher and its terms. Section 2 is that register, written for a reader who wants to know what is being relied on before reading what was concluded from it.

Three of the sources could not be reached from the machine this work was done on. Where that changed what could be delivered, the section says so and says what is standing in for the missing data.

The short version.

- OpenStreetMap carries the Irish network well at 38 kV and above and barely at all below it: 22% of the sub-110 kV line length, 2% of the distribution transformers.
- Four EirGrid planning cases become eight PyPSA networks. All are connected, all solve a DC power flow, all solve a linear optimisation with HiGHS.
- The conversion is checked against each case's own solved voltage angles. Correlation is 0.99 or better on seven of the eight networks.
- 87% of transmission buses were placed from OpenStreetMap names, and those placements sit a median of **14 metres** from EirGrid's own published station coordinates.
- Two methods that share no data beyond the network itself pick the same binding circuit: Letterkenny to Strabane, the 110 kV tie into Northern Ireland.
- Every hour of weather in the shipped networks was generated here. No hour of it ever happened, and Section 8 says exactly what that does and does not permit.

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1 What is in this repository

Two bodies of work share the checkout. They meet on one question: how much of Ireland's electricity system can be reconstructed from data anyone can download?

The coverage study asks whether OpenStreetMap can supply the distribution network for an energy-system model, and measures the answer instead of assuming one. It reads a national OpenStreetMap extract and EirGrid's own asset register, compares them against each other and against ESB Networks' published totals, and draws the result. Five modules do the work: `network.py`, `eirgrid.py`, `analysis.py`, `plots.py`, and a trio that builds a self-contained interactive map. Its conclusions are in `FINDINGS.md` and summarised in Section 3.

The model build takes the four PSS/E load-flow cases EirGrid publishes with its Ten Year Transmission Forecast Statement and turns them into something a study can run on. `psse.py` reads the file format, `pypsa_net.py` converts it, `geocode.py` adds geography, `northwest.py` extracts a region and checks it against a hand-built dataset, `profiles.py` and `synthetic.py` supply hours, and `build_kit.py` assembles a standalone kit.

`graph-workshop/` is a separate teaching project that happens to sit in the same checkout. It shares no code and no data with either half.

1.1 Running it

The coverage study rebuilds from a 392 MB download:

```
python network.py prepare      # download, verify, prefilter, cache boundaries
python analysis.py all         # the five measurements -> data/*.json
python eirgrid.py fetch        # EirGrid's own asset register -> GeoPackage
python plots.py all            # every map
```

The model build needs no download at all, because the study files and the OpenStreetMap substation list are both committed:

```
python geocode.py match data/TYTFS2024_studyfiles/*_V35.raw
python pypsa_net.py build      # -> data/pypsa/<case>_<scope>/
python pypsa_net.py verify     # connectivity, DC power flow, optimisation
python northwest.py extract    # the North-West region, both views
python synthetic.py build      # a seeded year of profiles
python build_kit.py            # -> participant-kit/
python -m pytest               # 215 tests
```

The participant kit is a directory that can be lifted out whole. Nothing in it imports anything from the build half:

```
cd participant-kit
python -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
python examples/a_dc_power_flow.py
```

Measured from a clean virtual environment, that takes 42 seconds and ends in a plotted map.

Every stage caches into `data/raw/` and skips itself when the cache is present, so the commands are safe to repeat and there is no order to remember.

2 Where the data comes from

Ten sources, of which seven are used, one is used only as a set of published totals to compare against, and two could not be reached.

source	what it is and how it arrives	read by
TYTFS 2024 study files	EirGrid and SONI's four published PSS/E v35 load-flow cases for the all-island system. Committed to data/TYTFS2024_studyfiles/, so nothing is fetched at run time [6].	psse.py
Transmission Development Plan 2024	EirGrid's own asset register, published as an anonymous ArcGIS feature service. Three layers: overhead line, underground cable, stations. Downloaded page by page and cached as a GeoPackage [3].	eirgrid.py
OpenStreetMap national extract	Geofabrik's island-wide extract, 392 MB, checksum verified, then filtered down to the 761,470 objects carrying a power tag [10, 15].	network.py
OpenStreetMap substations	Every named power=substation and power=plant on the island, from the Overpass API. 1,705 objects, committed as data/osm_substations.csv [15].	geocode.py
County boundaries	The 26 Republic of Ireland counties, from Ordnance Survey Ireland's statutory boundary service via Esri Ireland [16].	eirgrid.py
ESB Networks asset counts	Published headline figures for the distribution network. Used only as denominators, never as geometry [9].	analysis.py
Hand-built node list	Three spreadsheets supplied with the brief, describing the North-West region as 24 and then 15 nodes. Committed unchanged to data/handbuilt/.	northwest.py
ERA5 through Open-Meteo	Reanalysis weather, which would supply real hourly wind and solar [11, 13, 14]. Never fetched: the host is refused by this environment's egress policy.	profiles.py
EirGrid Smart Grid Dashboard	Published all-island demand and wind generation, which would supply the demand shape and validate the wind profiles. Never fetched , same reason.	profiles.py
Synthetic profiles	Generated in this repository from the case files alone, seeded with 42. This is the only source of hours in the shipped networks.	synthetic.py

Table 1: The source register. Everything downstream is built from these and from nothing else.

2.1 The TYTFS study files, in plain terms

EirGrid and SONI operate the transmission systems of the Republic and Northern Ireland. Once a year they publish a forecast of what that system will look like a decade out, and with it they release the working files behind it: four load-flow cases, each a snapshot of the whole island at

one moment, solved.

A case is a plain text file listing every busbar, every circuit, every transformer and every machine, with the electrical properties of each and the answer the operator’s own software produced for it. The four differ in two ways. Two are set on the 2024 network and two on the planned 2033 network, which has more of everything. Within each pair, one is a winter peak, when demand is highest, and one is a summer valley, when it is lowest.

case	condition	buses	branches	machines	running
WP2024	winter peak, 2024 network	2,025	1,129	597	87
SV2024	summer valley, 2024 network	1,979	1,120	558	18
WP2033	winter peak, 2033 network	2,457	1,296	847	710
SV2033	summer valley, 2033 network	2,457	1,296	847	45

Table 2: The four cases as the parser reads them. “Machines” is the generator register, which lists everything connected; “running” is how many of them the case actually dispatches. The gap between those two columns is the single easiest way to get a wrong answer out of these files, and Section 4 is about it.

Two things about the files are worth knowing before any number is quoted from them. They describe a forecast of how the system is planned to develop, and EirGrid publishes them for reference without warranty. And a case is one hour, so anything that asks when the network is constrained needs hours that the files do not contain.

2.2 EirGrid’s own asset register

The Transmission Development Plan web map is EirGrid publishing where its circuits and stations are. It is served as an ArcGIS feature service that anyone can query, in Irish Transverse Mercator, and `eirgrid.py` pages through three of its layers: existing overhead line, existing underground cable, existing stations.

One check that this really is the register: summing the geometry gives 5,348 km of overhead line and 774 km of cable, 6,122 km together, against EirGrid’s published figure of about 6,500 km.

One documented hole, which the aggregate hides. The 400 kV layer contains a single circuit, Moneypoint to Oldstreet, 103 km, which is the western leg. The 400 kV network continues east from Oldstreet to Dunstown and on to Woodland, and no circuit for those legs appears in the service at any voltage. The 110 kV and 220 kV layers can be taken as the register. The 400 kV layer cannot be taken as the whole 400 kV backbone. `eirgrid.py summary` prints that warning, and Section 6 repeats it where the same service is used as a validator.

2.3 OpenStreetMap, entered two different ways

OpenStreetMap is a map anyone can edit, so it records what volunteers have traced. That makes it a claim about mapping effort as much as a claim about the network, which is the whole subject of Section 3. This repository uses it twice, for two different purposes, through two different doors.

The bulk extract is a single 392 MB file covering the island, republished continuously by Geofabrik. `network.py` downloads it, checks it against Geofabrik’s published MD5, and streams it through a filter that keeps only objects carrying a `power` tag. The result is about 7 MB and holds 761,470 objects. Overpass was avoided here deliberately: one national file gives complete coverage in one pass, with no query timeouts, no rate limits and no result caps, and it makes the whole coverage study reproducible from one download.

Two practical notes for anyone repeating it. The download had to go over plain HTTP, because this session’s egress policy blocks HTTPS to Geofabrik and to every OpenStreetMap mirror tried, while allowing port 80; the checksum matched, which gives integrity checking

without authentication. And the prefiltering pass is there for memory and does not change the selection: the reader used downstream holds the whole node index in RAM and was killed on the full file with 15 GB available.

The substation query is small and specific. `geocode.py` asks Overpass for every named `power=substation` and `power=plant` on the island, which returns 1,705 objects: 1,288 substations and 417 plants. 1,270 of them carry a voltage tag and 367 are tagged at 110 kV or above. That list is committed to `data/osm_substations.csv`, so the bus matching in Section 6 reproduces without a network round trip.

OpenStreetMap data is under the Open Database Licence.

2.4 ESB Networks’ published totals

ESB Networks is the distribution system operator for the Republic. Its own website publishes headline asset counts: 150,000 km of overhead line, 22,000 km of underground cable, 2.1 million poles, 242,000 pole-mounted transformers, 21,680 ground-mounted substations and 571 primary stations [9]. None of that is geometry and none of it is used as geometry. It is used once, as the denominator that turns “OpenStreetMap contains 37,125 km of sub-110 kV line” into “OpenStreetMap contains about a fifth of it”.

The distribution network itself was not available for this work. Getting it means ESB’s CAD files and capacity heatmap, and Section 3 ends with an acceptance test to apply to them when they arrive.

2.5 The hand-built North-West dataset

Three spreadsheets came with the brief, describing the North-West of the country as a small node-and-edge model:

<code>nodes.xlsx</code>	the node list, in two blocks: 24 nodes with generation sites as nodes of their own, then a substation-only view of 15
<code>transmission.xlsx</code>	14 of the 15 stations, with the code, rating and length columns empty. A template with nothing filled in
<code>nodes_24hr.xlsx</code>	a half-hourly day for the 15 nodes. Its own README says it was fabricated, with NumPy seed 42

They are committed unchanged, because Section 7 is a line-by-line comparison against them and the reader has to be able to check it.

2.6 The two sources that could not be reached

`profiles.py` is a complete pipeline from reanalysis weather to a capacity factor per generator, with a validation step against EirGrid’s published wind output. It has never run. Both hosts are refused by this environment’s egress policy at the CONNECT, before any request is made:

```
archive-api.open-meteo.com:443 gateway answered 403 to CONNECT
www.smartgriddashboard.com:443 gateway answered 403 to CONNECT
cms.eirgrid.ie:443 gateway answered 403 to CONNECT
```

A 403 at CONNECT is an organisation policy denial. It was not retried and no mirror was sought, because routing around an egress policy is not this project’s call to make.

The consequence is stated plainly wherever it matters. There are no reanalysis profiles in this repository and no validation result. What stands in their place is described in Section 8, and it is generated data with a seed on it.

2.7 What is measured, what is inferred, and what was invented

The distinction runs through everything below, so it is worth one table.

measured	Every electrical property of the network: buses, circuits, impedances, ratings, transformers, the machine register, which buses carry load and how much. All read out of the TYTFS files. Station coordinates, read out of OpenStreetMap. Circuit and cable geometry, read out of EirGrid's register. Mapping coverage, measured off the OpenStreetMap extract.
inferred	What the twelve PSS/E rating columns mean (Section 5.2). Which technology each machine is, taken from bus-name prefixes, which leaves about a fifth labelled unknown . Which station a sub-110 kV bus belongs to, taken from least-reactance distance. Ratings for zero-impedance couplers, taken from what is attached to them.
invented	Every hour of weather. The shape of demand through the day. Marginal costs. The value put on unserved energy. These are stated wherever they are used and they are all in the exported CSVs, so a study can replace them.

2.8 How the pieces connect

```

Geofabrik extract ---> network.py ---> analysis.py ---> FINDINGS.md
                        |                               \--> plots.py --> maps
                        \--> extract_web_*.py --> build_web_map.py
EirGrid ArcGIS  -----> eirgrid.py -----^ and the geocoding cross-check

TYTFS .raw -----> psse.py -----> pypsa_net.py -----> data/pypsa/<case>_<scope>/
Overpass -----> geocode.py -----^
                                    |
                                    +-----> northwest.py --> the region
                                    +-----> synthetic.py --> the hours
                                    |
                                    build_kit.py --> participant-kit/

```

3 Part I. How much of the network does OpenStreetMap have?

The question came from an energy-system model that needed a distribution network and hoped to avoid asking the utility for one. The answer separates sharply by voltage, and the separation is the finding.

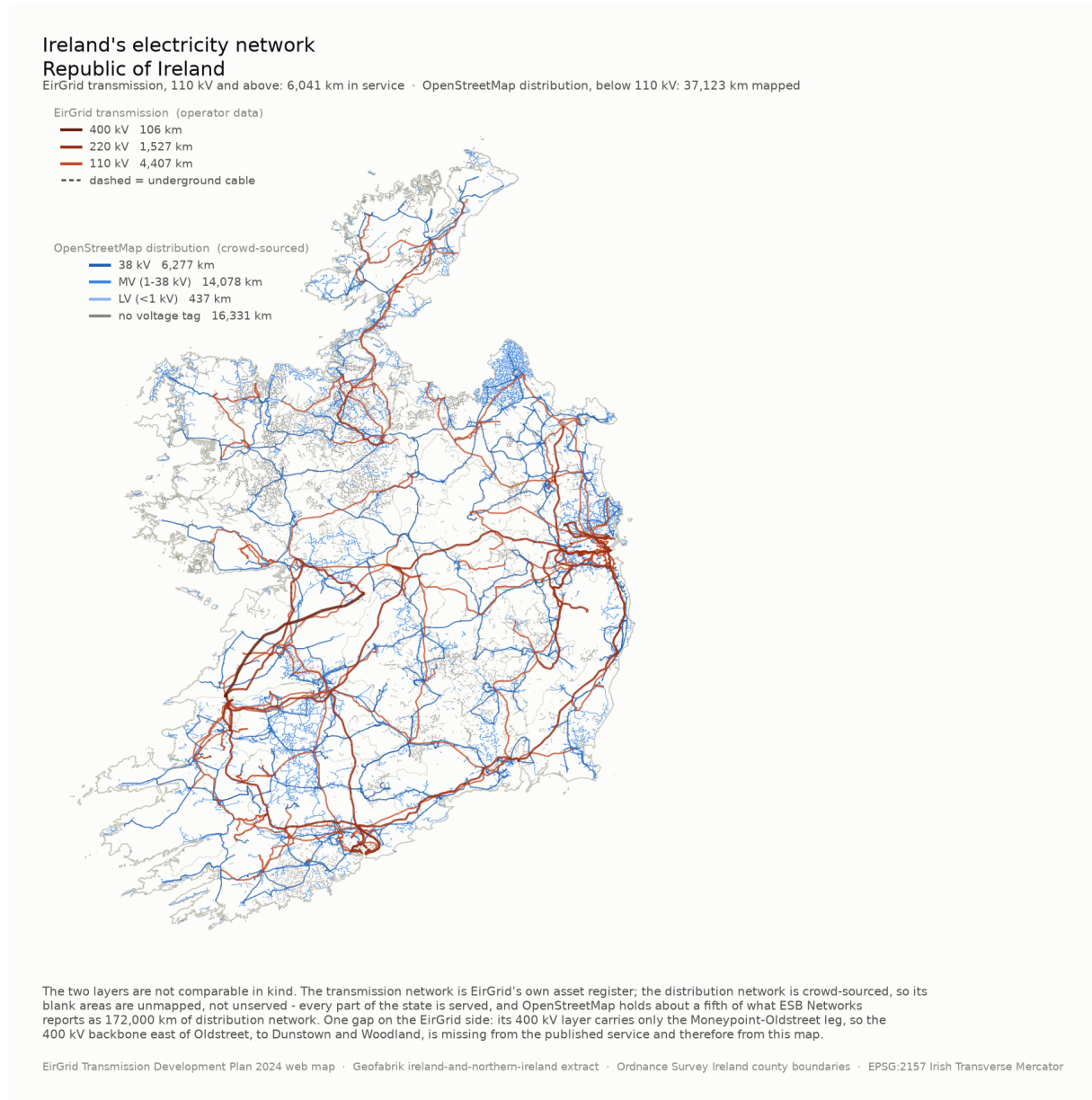


Figure 1: Both layers together. The warm lines are EirGrid's own register at 110 kV and above. The blue lines are what volunteers have traced below that. The colour families are deliberately far apart, because the two carry different kinds of claim: blank areas in the blue layer are unmapped rather than unserved.

3.1 One bug worth reporting on its own

The brief predicted that a line of the form `if gdf.get("voltage"):` would fail quietly. It does worse than that. When the column is absent the expression is `None` and the branch silently does nothing; when the column is present it raises, because the truth value of a pandas Series is ambiguous. It never behaves as intended.

That matters here because the reader used promotes only a fixed set of tags to columns and

leaves the rest in a JSON blob, and `voltage` is in the blob. A default read of County Kilkenny returns 16,078 line features with no voltage column, the branch skips, and the pipeline reports 0% voltage coverage. The true figure for Kilkenny is 48.4%. Nothing crashes, and the wrong answer confirms the hypothesis the study set out to test.

Code: `network.py → load_lines(), tag_series(); test_network.py` pins both cases

3.2 Coverage, against the operator’s own totals

Every OpenStreetMap power feature was attributed to a county and the 26 counties of the Republic summed. The attributed area comes to 70,296 km², within 0.03% of the state’s actual area, which is a reasonable check that the attribution works.

	OpenStreetMap	ESB Networks	share
Sub-110 kV line	37,125 km	172,000 km	21.6%
Wooden poles	470,731	2,100,000	22.4%
Underground cable	927 km	22,000 km	4.2%
MV/LV transformers	4,982	242,000	2.1%
Ground substations	987	21,680	4.6%

Table 3: The structure of this table is the finding. Line length and pole count agree at about 22%, which is two independent measures of the same thing and suggests roughly a fifth of the rural overhead network has been traced. Cable, transformers and substations sit at 2 to 5%, a factor of five worse. OpenStreetMap has a fifth of the wires and a fiftieth of the plant.

The transformer figure is the damaging one. Distribution substations are where a network-derived model would place its buses, and 2% of them are present.

A check that needs no external reference. Every transformer and substation OpenStreetMap maps has to be reached by conductor, whatever the real network looks like. A minimum spanning tree over the mapped assets is therefore a hard and very generous lower bound on the length required. Dublin City fails that bound against its own data: 874 mapped transformers and substations, 154 km of tree required, and 7.8 km of sub-38 kV line actually mapped. At least 95% of Dublin’s medium-voltage network is missing, and that follows without reference to ESB at all.

Code: `analysis.py → counties(), missing_cable()`

3.3 Topology: a scattering of segments below 38 kV

Run island-wide with no county clipping, because clipping cuts every line that crosses a boundary and would inflate the counts for exactly the layers that span counties:

layer	km	nodes	components	largest	share
220 kV and above	2,640	10,241	22	9,680	94.5%
110 kV	6,190	42,830	124	20,598	48.1%
38 kV	6,278	50,416	347	9,608	19.1%
38 kV and above	15,108	103,301	419	66,047	63.9%
MV, 1 to 38 kV	23,886	293,381	6,707	25,420	8.7%
LV, below 1 kV	619	17,495	4,682	149	0.9%

Table 4: From `data/national.json`. The upper layers behave like mapped infrastructure. The lower ones do not.

For context on the 6,707 medium-voltage components: ESB operates 571 primary stations nationally, and a radial network has roughly one component per primary busbar, so the true figure is of order 500 to 600. OpenStreetMap gives an order of magnitude more. Below 38 kV this is a collection of independently traced segments rather than a network. Someone walked a run of poles along one road, someone else walked another two townlands away, and the span between them was never traced.

Snapping does not fix it. If the fragmentation were imprecise noding, component count would collapse once the tolerance passed the noding error. Widening from 0.1 m to 25 m in Kilkenny, already far beyond any plausible tracing error, removes 18% of components. At 100 m the graph is being destroyed faster than it is being joined: node count falls by 68% while component count falls by 47%, and the largest component shrinks from 3,777 nodes to 1,361 as spurious merges rewire it. The missing spans are hundreds of metres of unmapped conductor, so no tolerance recovers them.

Code: `analysis.py` → `national()`, `analyse_area()`; `network.py` → `to_graph()`

3.4 Is it worth anything as a spatial prior?

For the line layer, no. Mapped sub-110 kV density across the 26 counties runs from 210 km per 1,000 km² in Westmeath to 1,450 in Monaghan, a factor of 6.9. The tempting reading is urban against rural, and it does not survive contact with the list: Monaghan and Westmeath are both small inland counties of similar character and differ by seven times, while County Dublin, which has the highest demand density in the state, sits third. What the ranking reproduces is where an active local mapper happened to work. Using mapped circuit length as a demand prior would import a sevenfold arbitrary regional distortion, which is worse than using no spatial information at all because it looks like data.

For the point layer the answer is more useful and comes with a caveat. Dublin City has 874 mapped transformers and substations in 119 km² against Kilkenny's 128 in 2,071 km², a density ratio of about 118 to 1 in the direction demand actually runs. Those urban substations get mapped because they are visible street furniture. The caveat is that this density partly measures mapper effort, mapper effort correlates with urbanity, and urbanity correlates with demand, so the point layer will look like a good demand prior whether or not it is one, and OpenStreetMap alone cannot separate the two.

3.5 The verdict, and what to do with it

Take the 38 kV-and-above layer. It is 15,108 km island-wide, 64% of its nodes sit in one component, and above 220 kV that rises to 94.5%. For placing regional nodes and constraining transfer between them it is adequate and it is free.

Do not use anything below 38 kV, as topology or as a demand prior. Go to ESB for that, and apply two cheap acceptance tests when the data arrives: does the medium-voltage layer resolve to something near 571 connected components nationally, and does its conductor length reach the transformer population? Both are the questions OpenStreetMap fails.

3.6 The interactive map

`data/ireland_distribution_map.html` is the same data as a map that opens from disk with no server. Three scripts build it: one reassembles the conductor runs and reduces polygon substations to points, one extracts the county and Northern Ireland outlines so the map still reads as Ireland with no tile server, and one inlines every byte of geometry, styling and script into a single file.

Two decisions carry the file size. Geometry is simplified in Irish Transverse Mercator at a tolerance chosen per layer, tight for the transmission layers a reader will zoom into and looser

for the untagged fragments they will not, then delta encoded against a grid about a metre across. Attributes are stored as arrays against one shared string pool, so “ESB Networks” is written once rather than eleven thousand times. Together those take the payload from roughly 90 MB of GeoJSON to 11 MB.

The default view is the layers that survived scrutiny, which is 38 kV and above plus the substations. Medium voltage, low voltage and the untagged remainder are one click away, with the reason stated beside them.

Code: `extract_web_data.py`, `extract_web_base.py`, `build_web_map.py` →

4 Part II. Reading the PSS/E cases

`psse.py` reads the four study files into pandas DataFrames and does nothing else. No per-unit conversion, no topology, no network object. Everything downstream of the file is another module's job.

Why not a library. PyPSA has no PSS/E importer, and the general-purpose readers that exist carry a solver or a whole network model behind them. The format is eight sections of comma-separated records with one genuinely awkward record type, so a dependency would cost more than it saves. The sections this repository does not need are skipped by name, so that each skip is a decision recorded in the code and not an accident of the file's order, and an unrecognised section name raises.

4.1 The format, as verified against the four files

Encoding is latin-1 and line endings are CRLF. A line beginning `0 /` delimits sections, and its comment names the section that follows it. Each section opens with one or more `@!` comment lines carrying the column headers. Records are comma-separated, character fields are single or double quoted, and anything after an unquoted `/` is a comment. Trailing fields may be omitted, so a short record is padded; a record longer than its layout raises, because that means the layout is wrong.

Two record types span several lines and are assembled into one flat row so that every section comes back as a rectangle. A transformer is four lines when it is two-winding and five when three-winding, and the third field on the first line says which. A two-terminal DC link is always three lines: link, rectifier, inverter.

The transformer assembly is checked with arithmetic. WP2024's transformer section holds 5,349 data lines, and the reader finds 1,201 two-winding and 109 three-winding records. $1201 \times 4 + 109 \times 5 = 5349$ exactly, with nothing left over. That is the failure mode that matters: one mis-sized record slides every record after it by a line, and the result still parses.

4.2 Validation, and the four checks that did not agree

Sixteen figures for WP2024 were supplied for the parser to reproduce. Twelve match exactly, including every voltage count, the branch classification, the in-service machine count and the dispatched megawatts. Four do not, and both causes were traced.

Three of them are record counts, each high by exactly one: buses, branches and loads. Each of those sections carries exactly one `@!` header line, and counting the lines between the two delimiters, instead of the records inside them gives the three expected figures precisely. The generator count is the tell: it agreed at 597, and the generator section has a header too, so a header-inclusive count there would have given 598.

The fourth is the load total, and it is the more interesting one. The expected 8,246 MW reproduces exactly as the sum over all 266 load records. What differs is which records to count. Thirty-four of them are switched out, they total 921.9 MW, every one carries the same identifier `MI`, and every one sits at a bus that already carries an in-service load record. They are an alternative representation of the same demand, deliberately disabled. The solved case settles it: in-service generation is 7,411.6 MW against in-service load of 7,324.5 MW, a gap of 1.2% that is the right size for losses, while against the full register generation would fall 835 MW short, and a solved load flow does not fail to serve 10% of its demand.

4.3 The in-service flag, and why it is the first thing to check

WP2024's generator section is a register of 597 machines totalling 18,158 MW, more than twice the island's winter peak. Filtering to in-service leaves 87 machines and 7,412 MW, which is a plausible winter-peak dispatch. The in-service fraction is not stable enough to approximate:

	records	in service	fraction
WP2024	597	87	15%
SV2024	558	18	3%
WP2033	847	710	84%
SV2033	847	45	5%

The reader itself does not filter, because a reader reads. Its `generators()` and `loads()` helpers filter by default, and `summary()` reports the in-service and register totals side by side so neither is hidden.

Code: `psse.py` $\rightarrow$ `read_raw()`, `read_all()`, `generators()`, `loads()`, `ac_branches()`, `summary()`

4.4 Three notes for what comes next

Branch counts are unfiltered by default. 679 AC circuits in WP2024 have both ends at 110 kV or above, of which 645 are in service. Both figures are defensible, and a downstream model should say which it used.

AC circuits never change voltage. All 679 join two buses at the same base kV, so classifying a circuit by voltage is unambiguous. Transformation lives entirely in the transformer section, and a test pins that.

The interconnectors are not all in one place. The two-terminal DC section holds only Moyle. East-West and Greenlink are modelled as buses with generator records instead. Anything that needs the island's cross-border exchange has to look in both places.

5 Part III. Turning a case into a PyPSA network

Six decisions matter, in the specific sense that getting one wrong produces a network that still solves and whose every downstream number is wrong. Each is below with the measurement behind it.

5.1 The per-unit base: 100 MVA, read out of the header

Every case opens with `SBASE = 100.00` and `BASFRQ = 50`. The build asserts it and refuses a case on any other base, because a case on 200 MVA would produce a network that solves and whose every impedance is out by exactly two.

PSS/E gives impedances per-unit on that base and on the bus's kV. PyPSA wants ohms for a line and per-unit on its own rating for a transformer, so there are two conversions:

	PyPSA units	conversion
<code>Line.r</code> , <code>Line.x</code>	ohms	$R_{pu} \times kV^2 / 100$
<code>Line.b</code>	siemens	$B_{pu} \times 100 / kV^2$
<code>Transformer.r</code> , <code>.x</code>	pu on its own <code>s_nom</code>	$X_{pu} \times s_nom / 100$

Transformer records carry their own convention flags and all four cases use the same ones, with two exceptions on a different impedance base that are rebased explicitly. A third convention, load loss in watts, does not occur, and encountering it raises.

Code: `pypsa_net.py` → `_ohms()`, `_siemens()`, `_z_on_system_base()`

5.2 RATE1 is taken as the rating, by inference from the data

PSS/E v35 carries up to twelve rating slots per branch. RATE1, RATE2 and RATE3 are populated here and RATE4 to RATE12 are zero everywhere.

A caveat this report owes the reader. Siemens' own manual, which would say what the format fixes about their meaning, is not publicly available. Open-source readers document the older three-rating layout without settling the semantics of the twelve-rating one [17]. What follows is an inference from the data.

Two facts, measured across all four cases and re-derived for this draft:

- RATE1 == RATE2 in **every one of the 4,841 branch records** across the four cases: 1,129 in WP2024, 1,120 in SV2024 and 1,296 in each 2033 case. Exactly, not approximately.
- RATE3/RATE1 is 1.0 or 1.1 in the two winter cases. The two summer cases add a third value, 1.10021, on 97 records, which is a rating that was rounded before it was written.

The natural reading is the normal, long-term-emergency and short-term-emergency convention, with the long-term rating set equal to the normal one and the short-term rating at 110%. The magnitudes agree: the median RATE1 by voltage in WP2024 implies 659 A at 110 kV and 1,475 A at 220 kV, which are continuous circuit ratings.

`s_nom` therefore takes RATE1, the most conservative of the three. If that reading is wrong, the direction of the error is known: `s_nom` would be too small and every loading figure here too high. All three columns are exported, so a contingency study can pick a different one.

Code: `pypsa_net.py` → `CONTINUOUS_RATE`; `test_pypsa_net.py` → `test_rate2_equals_rate1`

5.3 The 18 unrated branches, and what they turned out to be

Eighteen AC branches at 110 kV and above carry no rating in WP2024: nine at RATE1 = 9999 and nine at RATE1 = 0. Zero is the more dangerous of the two because it looks like a number, and putting it into `s_nom` makes the element unusable and the optimisation infeasible.

Looking at what they are settles the treatment. All eighteen have $R = 0$, $|X| \leq 0.01$, $B = 0$ and zero length, as do all twenty in each 2033 case. Every one is a station coupler: a busbar section, a capacitor or reactor stub, an SVC, a generator terminal tie. None is a circuit whose rating was forgotten, and a test pins that, because if it stops being true the rule below is the wrong rule.

The rule has two parts. A coupler is bounded by what is attached to it, so its `s_nom` becomes the smaller of the two ends' summed rated incident capacity, which gives it a real limit instead of a placeholder. Everything else keeps the file's 9999.

That second half was arrived at by trying the other rule. Bounding the sub-110 kV distribution transformers the same way produced limits that bind on the case's own solved flows on 13 transformers, the worst at 156% of the bound it had been given, and made the full network's optimisation infeasible. There is no honest way to infer a rating for a 110/38 kV transformer from the feeders leaving the 38 kV bus, because load taps off that bus directly and the feeders bound nothing.

Code: `pypsa_net.py` → `_is_coupler()`, `_fill_unrated()`, `rating_exceptions()`

5.4 The sub-110 kV tail is reduced, not deleted

The networks come in two scopes, chosen by a `min_kv` flag. The measurement that decides how the floor works: in WP2024, 100% of in-service generation and 99.8% of in-service demand sits below 110 kV. A transmission network built by dropping those buses is empty.

So each removed bus is aggregated into the retained bus it reaches by the least-reactance path, and its generators and loads move with it. Paths through another retained bus do not count, because a load reached only by passing through a second transmission station belongs to that station. Where a dropped component touches more than one retained bus, which 137 of WP2024's 518 sub-threshold components do, the least-reactance one wins and both the runner-up and the margin are recorded, so the arbitrariness is visible.

```
model = pypsa_net.build(case, min_kv=pypsa_net.TRANSMISSION_KV) # 110 kV+
model = pypsa_net.build(case, min_kv=0.0)                       # everything
```

What the floor costs, measured. Both scopes were built, a DC power flow run on each, and the flows compared on the 645 circuits they share: mean absolute difference 2.39 MW, 93 circuits differing by more than 1 MW, worst 53.5 MW on the Ballylumford to Eden 110 kV circuit. The reduction is exact wherever the sub-threshold network is radial, which is most of it.

Three-winding transformers are why a naive filter would have failed even for topology. Only 11 two-winding transformers have both ends at transmission voltage; the 220/110 kV transformation is almost entirely in 97 three-winding units whose third winding sits at 10.5 to 22 kV. Drop those and the 220 kV and 110 kV networks come apart. Each becomes the textbook star equivalent: a new bus at the star point and one two-winding transformer per winding. Winding one's leg reactance comes out negative in 24 of WP2024's 109 records, which is what an autotransformer's star equivalent does, and it is carried through as it stands, unclamped.

Code: `pypsa_net.py` → `_retained()`, `_aggregate_to()`, `_star_equivalent()`, `build()`

5.5 The three interconnectors, modelled two ways in the file

Moyle appears in the two-terminal DC section, twice, once per pole, running from a bus named SCOTLAND that has no AC branch and no transformer of its own. East-West and Greenlink appear instead as buses with generator records and no far terminal at all.

All three become PyPSA Link components, so that the exchange is a controllable flow on a branch. That matters because a generator cannot run backwards and an interconnector can. The two missing far terminals are created, since a link needs two buses, and each carries an import generator priced above every domestic carrier and a separate export sink. One

bidirectional generator would be simpler and wrong: a machine with a positive marginal cost and negative output is paid to run backwards, and the first version of this exported 1,223 MW in the optimisation for exactly that reason.

All three are real links. Moyle runs from Scotland to Co. Antrim and has been in service since 2001. The East-West Interconnector runs from Portan in Co. Meath to Shotton in Wales, 500 MW, in service since 2012. Greenlink runs from Great Island in Co. Wexford to Pembroke and entered commercial operation in 2025 [4]. The file names the last of these GREENLINK; OpenStreetMap names the converter station after the townland, Campile, and it sits 400 m from the Great Island 220 kV substation, which is why `geocode.py` carries that as a hand-written alias with its reasoning attached.

Moyle's efficiency is computed from the record's own DC resistance and scheduled voltage, which gives 0.84% loss per pole at rated power. That is optimistic, because converter losses are not in the record and were not invented. East-West and Greenlink have no loss data at all and are left lossless, which is said here so that it is visible.

Code: `pypsa_net.py` → `INTERCONNECTORS`, `_add_links()`, `_moyle_efficiency()`

5.6 The reference bus, and the bug that found it

Each case names its own reference buses: five carry the PSS/E type code for a swing bus. The rule is to honour them where they survive the voltage floor, hand the role to whatever absorbed a swing bus that was aggregated away, and otherwise give it to the bus carrying the most in-service generation. Every choice and the rule that made it goes into `reports/slack.csv`.

This one is not cosmetic. Determining the network topology makes PyPSA choose a slack of its own, the first generator in each sub-network, and in WP2033 that is a 21 MW hydro unit at Ardnacrusha behind its own step-up transformer. A lossless DC flow has nowhere to put the case's 198 MW gap between generation and demand except the reference, so that one machine absorbed all of it: 856 MW through a 21 MW step-up, 172° across it, and the whole network's angles thrown out by up to 100° .

The fix is to assign the reference as a whole column after the topology has been determined, instead of label by label:

```
# Assigning per-label leaves PyPSA's own choice in place. Build the whole
# Series and set it as a column, after determine_network_topology().
control = pd.Series("PQ", index=n.generators.index)
control[chosen] = "Slack"
n.generators["control"] = control.values
```

Angle correlation against the case's own solved angles: **0.90 before, 0.99 after**.

One consequence is worth stating. The reference still absorbs the case's whole imbalance, 87 MW in WP2024 and 198 MW in WP2033, because that gap is AC losses and a lossless model has nowhere else to spend it. The reference machine's own terminal angle is therefore not comparable with the case's, which is why the angle check below centres on the median.

Code: `pypsa_net.py` → `_slack_buses()`, `build()`

5.7 Verification

`python pypsa_net.py verify` runs all four cases at both scopes. The network object and its linear power flow are PyPSA's [2]; the optimisation is solved with HiGHS [12].

case	scope	connected	DC PF	angle corr	angle sd	optimisation
WP2024	transmission	yes	solved	0.9961	0.93°	optimal
WP2024	full	yes	solved	0.9915	1.19°	optimal
SV2024	transmission	yes	solved	0.9987	0.36°	optimal
SV2024	full	yes	solved	0.9963	0.61°	optimal
WP2033	transmission	yes	solved	0.9907	1.94°	optimal
WP2033	full	yes	solved	0.9048	4.35°	optimal
SV2033	transmission	yes	solved	0.9977	0.25°	optimal
SV2033	full	yes	solved	0.9939	0.40°	optimal

Table 5: All eight networks. The angle check is the real verification of the conversion: the raw file records the solved voltage angle at every bus, so the result can be checked against the case rather than against itself. If the reactances, the star equivalents or the phase-shift signs were wrong, a linear flow’s angles would not track them. This is a linear flow against an AC solution, so the expectation is that the angles track to within a degree or two.

Two further results from the same run are reported as they came out. WP2033’s own dispatch, evaluated as a lossless DC flow, exceeds RATE1 on four transmission circuits, the worst at twice its rating on the Killonan to Singland 110 kV circuit. All four carry a RATE1 of their own, so none of them is an element this module gave a bound to, and the count of inferred limits is zero in every case at both scopes.

What is deliberately not carried. The reader skips the fixed and switched shunt sections, so this network has no reactive compensation and a full AC power flow will not reproduce the case’s voltages. DC power flow and linear optimisation ignore reactive power and are unaffected. The capacitor and reactor buses remain as dead ends carrying no injection, which is harmless in a linear flow and is where their coordinates come from in the next section.

Code: `pypsa_net.py` → `verify()`, `angle_check()`, `_connected_with_links()`

5.8 What gets written out

One directory per case per scope, in PyPSA’s own CSV folder format, so that handing the directory to PyPSA reads it back with nothing lost. Names are traceable to the record they came from: a bus is its PSS/E number, a line is I-J-CKT, a three-winding star bus is `star:I-J-K-CKT` with its legs suffixed.

Beside each network is a `reports/` directory whose `README.csv` lists what each table answers: every bus below the floor and where its load and generation went, every element the file left unrated and the bound it was given, every element dropped because an end was below the floor, the reference bus of each sub-network and why, the interconnectors and which representation each came from, and one row per bus recording how it was geocoded.

6 Part IV. Putting the buses on a map

A PSS/E bus record is a number, a twelve-character name and a base voltage. Where the station is, is not in the file. Every map, distance and spatial join a downstream model wants has to come from somewhere else, and those twelve characters are the only thing to join on.

`geocode.py` matches them against the 1,705 named OpenStreetMap substations and plants described in Section 2. Buses are grouped into stations first, because a station's busbar sections are one place. **Nothing is interpolated, averaged or guessed:** a station that cannot be matched gets no coordinate and is reported by name, with the candidate it rejected and the reason.

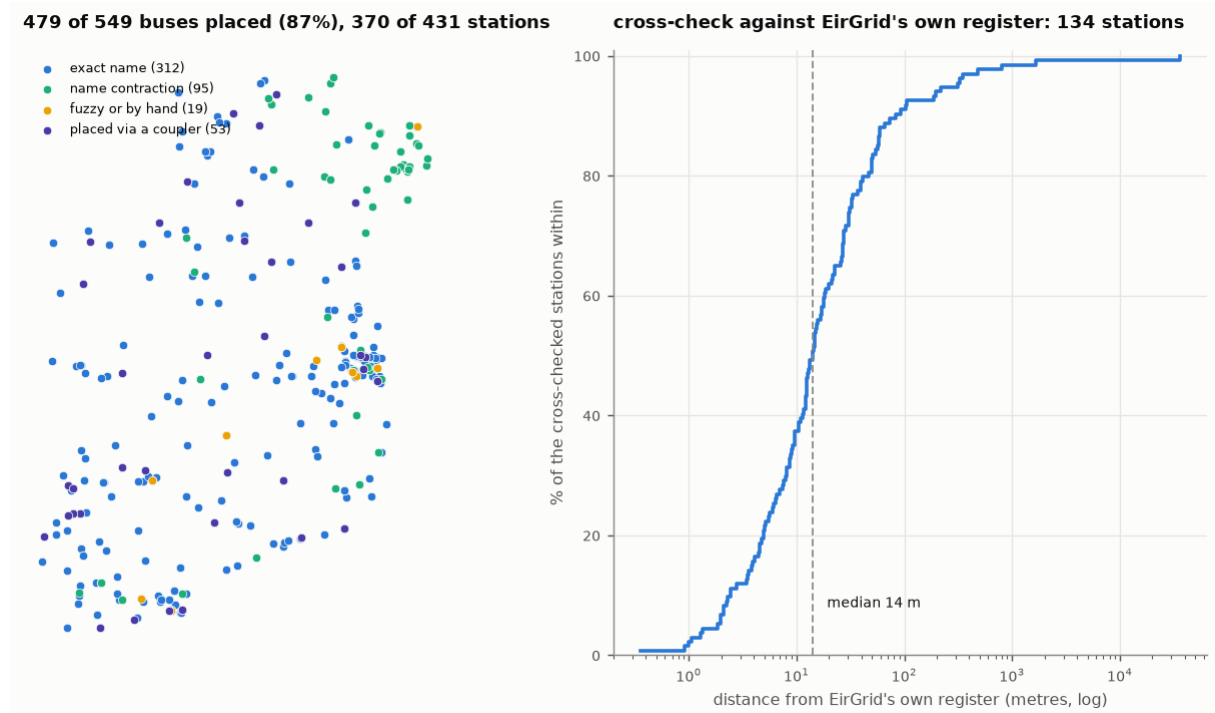


Figure 2: Left: where the WP2024 transmission buses landed, coloured by how each was matched. Right: the cross-check against the station layer of EirGrid's own Transmission Development Plan service [3]. 134 of the matched stations appear in both, at a median separation of **14 metres**, which is the difference between one corner of a substation and another. The 95th percentile is 252 metres.

method	buses	what it means
exact	309	normalised names equal, voltage tag agrees
coupled	53	joined to a placed bus by a coupler or its own transformer
ni-code	60	Northern Ireland contraction resolved by subsequence
truncated	19	the twelve-character name is a prefix of exactly one candidate
fuzzy	13	token-sort score ≥ 92 with ≥ 6 points over the runner-up
prefix	10	six characters or more, matching exactly one candidate
alias	6	resolved by hand, with the reasoning recorded
ni-site	6	the same Northern Ireland site at its other voltage class
exact-name	3	names equal, candidate carries no voltage tag
unplaced	70	every one named, with its rejected candidate

Table 6: 479 of WP2024’s 549 buses placed, which is 87%. Counts re-derived for this draft from data/pypsa/geocoding/. Voltage is corroboration rather than a key: OpenStreetMap’s voltage tag is present on 1,270 of the island’s 1,705 named power objects, so agreement raises confidence, absence does not lower it, and a contradiction rejects the match.

6.1 Two rules that do most of the work

The Northern Ireland contraction. Buses there are named with a four or five letter contraction and a digit: ANTR1A, BAFD2-, MAGF2-. The letters are a subsequence of the station’s name rather than a prefix, so BAME is Ballymena, BNCH is Ballynachinch and MAGF is Magherafelt. No amount of fuzzy string matching reaches those, so they are matched by subsequence and scored on how tightly the letters sit. The trailing digit is a voltage class, 1 for 110 kV and 2 for 275 kV, which holds for every pair the cases contain.

Coupling. A capacitor, reactor, SVC or phase-shifter bus carries a name of its own, such as MOB_CAP or ENNK_PST, which no name match will ever reach because it is not the name of a place. Two things put two buses inside the same fence: a zero-impedance branch, and a transformer between two buses both at 110 kV or above, which is the station’s own transformer standing in its yard. Together those place 48 stations that string matching would have missed.

6.2 The three hard cases

Cathleen’s Fall appears in the study files as CATH_FALL and CATH FALL, because the name does not fit in twelve characters. OpenStreetMap spells it Cathleen’s Fall, without the second ‘a’. Neither spelling reaches the other by truncation or by fuzzy matching, so it is in the alias table with that reasoning written beside it.

Clogher is four separate 110 kV buses in the file and one substation in OpenStreetMap. The four are busbar sections of one site, which is exactly what the zero-impedance coupler between two of them says, so they are grouped before matching and share one coordinate.

Clady is absent from the study files at any voltage, and that is not a naming problem. OpenStreetMap has a Clady 38 kV substation and a Clady hydroelectric station in Gweedore, both tagged at 38 kV. EirGrid’s own 110 kV-and-above register does not contain it either. A 38 kV station sits below the transmission model’s floor, and Clady’s four megawatts do not earn a bus of their own.

6.3 The 70 unplaced buses

Every one is listed with its method, its best rejected candidate and the score. Fifty-five scored nothing above the threshold, and most are genuinely not in OpenStreetMap: the island has 367 substations tagged at 110 kV or above against 431 stations in the case, so a residue is expected. Four are ambiguous, where two candidates cannot be separated: DRUM1 sits equally

tightly inside Drumnakelly and Drumlins Park Wind Farm. Three are rejected because the only name match is at the wrong voltage class. One is deliberate: SCOTLAND is the GB end of Moyle, a boundary bus and not a place in Ireland, so it is left without a coordinate instead of being put at Auchencrosh.

The 2033 cases place 78% against 87%. The difference is what the forecast statement exists for: stations that do not exist yet are not in OpenStreetMap.

6.4 The one disagreement worth reporting

Of the 134 cross-checked stations, one is more than 5 km out. TANDRAGEE sits 36.3 km away, and EirGrid’s register is the one that is wrong: it puts the station at the Republic’s end of the Louth to Tandragee 275 kV interconnector, in Co. Louth, while the substation is in Co. Armagh where OpenStreetMap has it.

A caveat on the validator. The same feature service is known to be incomplete on its line layers, as Section 2 sets out. The station layer shows no such gap and the cross-check is station to station, but a validator with a documented hole in it should be named as one.

Code: `geocode.py` → `normalise()`, `_subsequence_span()`, `ALIASES`, `match()`, `crosscheck()`

6.5 The stated line lengths are real, and here is the check

An earlier draft of this report listed the line lengths as placeholders. Checking it instead of repeating it, and re-deriving the figures for this draft directly from the raw files:

- **83% of transmission branches carry a non-zero length**, 561 of 679 in WP2024, with a median of 11.7 km and a longest of 208.5 km.
- Of the 118 that do not, **111 are couplers and stubs** with $|X| \leq 0.01$, where a zero length is the correct value. Seven are real circuits missing a length, and six of those seven are in Northern Ireland.

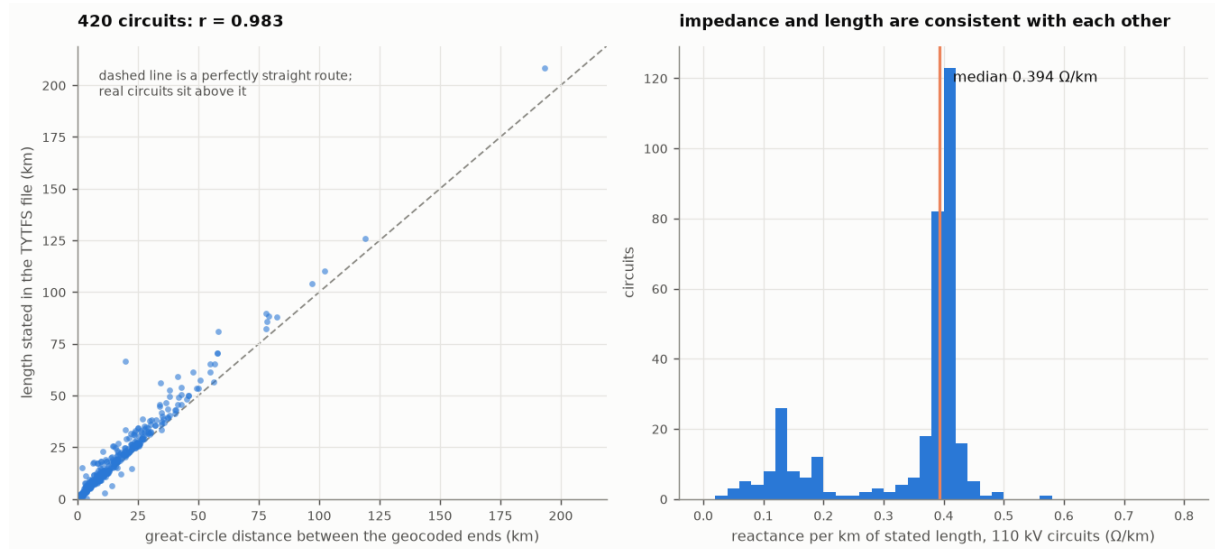


Figure 3: Left: for the 420 circuits with a stated length and both ends independently geocoded, the stated length against the great-circle distance between the ends. $r = 0.983$, median ratio 1.17, which is what a real route does against a straight line. Two datasets that share no source agreeing this closely checks the geocoding as much as it checks the lengths. Right: reactance per km of stated length at 110 kV.

The right-hand panel is bimodal, and the split is physical. The 347 circuits above 0.25 Ω/km have a median of 0.394 Ω/km and a median shunt susceptance of 0.00035 pu/km. The 117

below have a median of $0.13 \Omega/\text{km}$ and $0.011 \text{ pu}/\text{km}$, which is thirty times the charging. High capacitance with low reactance per km is the signature of underground cable, and the names in that cluster are Ringsend to Whitebank, Galway to Salthill, Marina to Trabeg: urban and coastal, where cable is what gets laid.

Two things follow. The lengths are usable, with the seven gaps named. And the participant kit's fallback of $0.4 \Omega/\text{km}$ for a new circuit, which an earlier draft called a rule of thumb, is within 2% of the median of this dataset's own overhead-line population.

7 Part V. The North-West region

EirGrid and SONI run a Wind Dispatch Tool in their control centres, which applies megawatt limits to individual wind and solar farms or to groups of them associated with a particular network constraint. The group definitions are published [7]. The region studied here is Donegal, Sligo and north Mayo: heavily wind-dominated, behind a narrow export path, and identified in the brief as constraint groups 1 to 3.

What the published groups are, and why the numbering matters. The February 2024 overview lists the North West groups as Letterkenny A1 busbar flows, Letterkenny A2 busbar flows, Carrick-on-Shannon to Arva, two outage-driven groups covering Sliabh Bawn to Lanesboro and Binbane to Cathaleen’s Fall, and Corderry to Srananagh, which was retired after that circuit was uprated. The 2020 version numbered them differently: group 1 was Letterkenny to Clogher and group 2 Corderry to Srananagh [5, 7].

Two consequences. The station set studied here is clearly the right region: Letterkenny, Clogher, Binbane, Cathaleen’s Fall, Corderry and Srananagh all appear in EirGrid’s own group definitions. But the numbering is not stable across versions, so the phrase “groups 1 to 3” does not by itself pin down a fixed set of stations, and nothing here should be read as reproducing a particular numbered group.

Neither overview document could be retrieved from this environment, since the hosts are refused by its egress policy, so those group names come from a search index rather than from the documents. They should be checked against the PDFs before being quoted onward.

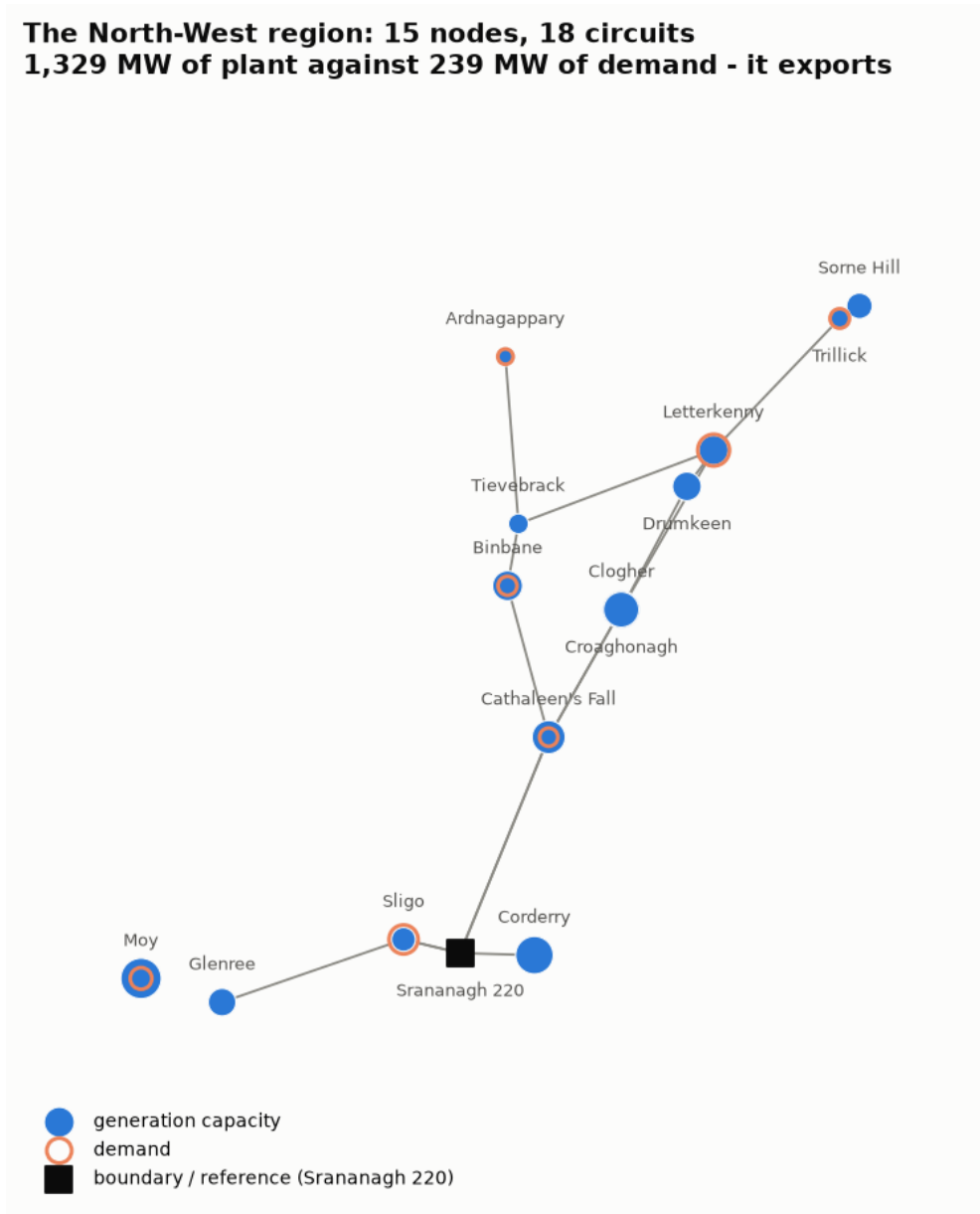


Figure 4: The region as the kit ships it. Dot area is plant capacity, the ring is demand, the square is Srananagh 220 kV, which is both the boundary and the reference. 1,329 MW of plant against 239 MW of demand, so it exports, and the question is whether the wires can carry it.

7.1 Two views, because the two datasets draw a station differently

The hand-built dataset folds a generator's connection point into the substation it hangs off. The study files give several of them 110 kV buses of their own. Both views are therefore produced, and the mapping between them is one table in the code:

```
STATIONS = (
    Station("Srananagh 220", (5042,), 220.0, note="the region's export path"),
    Station("Srananagh 110", (5041,), 110.0, parent="Srananagh 220"),
    Station("Cathaleen's Fall", (1701, 17010, 17061), 110.0),
    Station("Cliff", (1761,), 110.0, parent="Cathaleen's Fall"),
    ...
)
```

The native view is 24 stations, matching the hand-built 24-node block one for one with a single substitution: the study files have no Clady, and they split Srananagh into a 110 kV and a

220 kV busbar that the spreadsheet holds as one node. The aggregated view folds nine of those into their parents and gives the 15 nodes the spreadsheet has.

The region is listed by name instead of being derived, for two reasons. A constraint group is a published grouping, and not a property of the network that anyone could compute. And the region is not the radial island behind Srananagh that a purely topological definition would find: removing the Srananagh transformer leaves every bus in the region still connected, through Corderry and Arigna towards Flagford, and through Letterkenny and Corraclassy into Northern Ireland.

Code: `northwest.py` → `STATIONS`, `station_table()`, `bus_map()`, `extract()`

7.2 The circuit count

The brief said 16 circuits among the 15 stations. Reproduced independently: **16 routes, carried by 19 circuits**. Both numbers are right and they answer different questions. A route is a pair of stations and a circuit is a wire, and three of these routes are double circuits whose two wires land on different busbars at one or both ends, which is exactly why they collapse to one edge in a node-and-edge list. The native 24-station view has 25 routes and 29 circuits.

7.3 What folding to 15 nodes costs

Same impedances, same injections, a different reference bus: a DC flow has to put the same megawatts on every circuit as the full network does. That makes the fold a check with a number attached.

what is folded	worst disagreement against the full network
nothing (24-station native view)	0.0099 MW
all nine folds (15-station view)	21.59 MW
all but Srananagh 110	1.74 MW
all but Srananagh 110, Lenalea, Cunghill	0.15 MW

Table 7: Almost all of the error is one fold. Merging Srananagh’s two busbars short-circuits the 250 MVA transformer between them, whose 0.064 pu of reactance is a large fraction of the region’s total, and that redistributes the two Cathaleen’s Fall to Srananagh circuits by 21.6 MW each.

The recommendation is one node. Split Srananagh back into 110 and 220 kV with the transformer between them, giving 16 nodes instead of 15, and the flow error falls from 21.6 MW to 1.7. That is a cheap change to a dataset that is otherwise sound.

Code: `northwest.py` → `compare_with_full()`, `extract()`, `verify()`

7.4 What the reconciliation found in the hand-built data

The wind capacities are right. Four of the hand-built figures agree with the study files to within 1%: Mulreavy at 95 against 95.25, Meentycat 85 against 84.96, Cunghill 35 against 34.80, Lenalea 30 against 30.50. These are the same numbers rounded.

The hydro capacities are all identically 38, and none of them matches. Cathaleen’s Fall, Cliff and Clady are given 38 MW each. Cathaleen’s Fall alone is two machines of 22.5 and 23 MW in the study files; Cliff is two of 10; Clady’s real rating is about 4.2 MW. All three plants connect at 38 kV, 278 buses in WP2024 carry that voltage, and Cathaleen’s Fall has a bus named HY_CATH FALL sitting at exactly 38 kV. A connection voltage entered into a capacity column is the reading that fits, and it is worth checking before the 76 MW the spreadsheet puts at Cathaleen’s Fall is used for anything.

The two blocks of the spreadsheet disagree by exactly 35 MW, which is Cunghill. The 24-node block assigns it to Sligo and the 15-node block gives Sligo no supply at all.

The supply set is a correct subset of a larger one. Every plant it names is in the study files at the station it says. The study files carry 1,155 MW of connected capacity across those 15 stations against the spreadsheet’s 425 MW, and the gap sits almost entirely at stations the spreadsheet gives zero supply: Croaghonagh at 139 MW, Moy at 190, Corderry at 145, Binbane at 75. So the region is more export-dominated than the hand-built data says, at 5.4 to 1 against the hand-built 2.7 to 1.

Demand agrees in total and not in distribution. 160 MW against 213 MW is agreement for a nominal figure against a winter peak. But nine of the fifteen nodes carry demand in the spreadsheet that the study files put at zero, because they are transmission stations with no 38 kV load busbar beneath them, while Letterkenny carries 66 MW in the files against 15 in the spreadsheet. The pattern is a nominal per-station allocation of 5, 10, 15 or 20 MW against the case’s actual load at the six stations that have any.

The 15-node region is a 2024 region. In both 2033 cases it comes apart, and not as a modelling artefact: the 2033 network rebuilds the Mayo 110 kV system. The Glenree to Moy circuit has no record at all, a new station called Firlough sits between them instead, and Moy gains ties to two other new stations. Firlough is not in the node set, so Moy falls out of the region and the extract has two components. `northwest.py verify` reports that as a failure, which is the right answer.

7.5 Where the constraint actually is

`python northwest.py windy -capacity-factor X` puts the region’s capacity to work at X and re-solves *the whole transmission network*. Not the extract: an extract with five of its six ties pinned sends every extra megawatt out through the sixth and gives an answer that is both large and meaningless.

capacity factor	net export	worst tie	Srananagh	worst internal
10%	−97 MW (importing)	Letterkenny/Strabane 0.36×	−85 MW	0.40×
20%	18 MW	Letterkenny/Strabane 0.71×	−62 MW	0.48×
30%	134 MW	Letterkenny/Strabane 1.05×	−39 MW	0.56×
60%	480 MW	Letterkenny/Strabane 2.09×	+31 MW	0.80×
100%	942 MW	Letterkenny/Strabane 3.48×	+123 MW	1.11×

Table 8: The binding constraint is the Letterkenny to Strabane 110 kV tie into Northern Ireland, circuit 3581-89516-1, and it binds at about 29% of connected capacity. Srananagh does not reverse into export until about 45%. **The rating is scenario-dependent:** RATE1 on this circuit is 93 MVA in WP2024, the case this sweep runs on, and 80, 105 and 123 MVA in SV2024, SV2033 and WP2033. All four were re-read from the files for this draft. Quoting one number for it across scenarios is wrong.

So Srananagh 220 is the right place to slack the extraction and the wrong place to look for the constraint. Two caveats are real and load-bearing: nothing outside the region is re-dispatched, and the system reference is Turlough Hill, far away. A study that cares should redo this with the Northern Ireland network dispatched rather than held fixed.

7.6 Both north-western cross-border ties sit behind a phase shifter

This came out of chasing a numerical discrepancy in the participant kit (Section 9) and is worth stating on its own. Reading the transformer records directly, and re-reading them for this draft:

from	to	rating	shift, winter	shift, summer
89510 STRABANE (NI)	89515 STRA_PST	125 MVA	17°	12°
79010 ENNISKIL (NI)	79015 ENNK_PST	125 MVA	3°	3°

Table 9: The only two phase-shifting transformers with both ends at transmission voltage, present in all four cases. PST is the file’s own naming. The Strabane angle differs between the winter and summer cases, which an earlier draft of this report missed by quoting 17° for all four.

The circuit this report keeps returning to, 3581-89516-1, runs from Letterkenny to bus 89516, which sits behind 89515 and therefore behind the phase shifter. The other north-western cross-border tie, Corraclassy to Enniskillen, has the same arrangement. So the binding element sits directly behind the device a system operator installs precisely to control flow on a tie.

This model holds both taps fixed. That is a sharper statement of the limitation than saying SONI would redispatch around it: there is a specific, identified control device on the constrained path whose action is not modelled here.

Code: `psse.py` → `read_raw()` → `case.transformer`, field `ANG1`

7.7 None of the four published cases runs the fleet

Worth stating before any of the above is read as a description of the system as operated. Between 3% and 21% of the region’s machines are dispatched in any of the four cases, because these are peak and valley security studies and a security study does not credit wind. Two of the four cases export, by 23 and 35 MW, against more than a gigawatt of connected capacity. The export the Wind Dispatch Tool exists to manage does not appear in any of them, which is why the capacity-factor sweep above exists.

8 Part VI. Turning four snapshots into hours

Each case is one hour with a solved dispatch. Anything that asks when the network is constrained needs a year, and the year has to have weather in it.

8.1 The pipeline that could not run

`profiles.py` is a complete pipeline against Open-Meteo's historical weather API [14], which serves ECMWF's ERA5 reanalysis [11] on a 0.25° grid and the higher-resolution ERA5-Land [13] on a 0.1° grid, without a key and under CC BY 4.0.

Reanalysis is the right source for one specific reason. Calm anticyclones and storm fronts are hundreds of kilometres across and cross Ireland in hours, so every wind farm in Donegal reaches rated output within the same few hours and falls to zero within the same few hours. That joint behaviour is what loads a constrained tie, and a reanalysis has it because it is a reconstruction of what the atmosphere actually did.

The module does batched requests, caches on disk by grid cell and year so nothing is fetched twice, computes an hourly shear exponent from the two ERA5 wind levels instead of assuming one, smooths a stated turbine power curve into a farm curve by Gaussian convolution after Staffell and Pfenninger [21], runs a panel model using the Erbs diffuse-fraction correlation [8], allocates the island demand across buses in proportion to each bus's own load, and validates the modelled fleet against EirGrid's published all-island wind series.

It has never run, for the reason given in Section 2.6. Both hosts are refused at the CONNECT.

So there are no reanalysis profiles in this repository and no validation result. What exists instead is the pipeline, tested, with the fetch as the only untested step, and every path that cannot get real data raises instead of falling back on something invented. That behaviour is itself tested:

```
test_loading_a_cell_that_was_never_fetched_raises
test_a_partial_year_is_not_cached_as_a_complete_one
test_a_response_missing_a_variable_is_refused
test_an_unreadable_eirgrid_file_raises_rather_than_guessing
```

The brief was explicit that the validation is the only thing standing between a real profile and a plausible-looking fabrication. Shipping a generated year under the ERA5 label, however carefully footnoted, would be that fabrication.

Two ways to finish it. Allowlist the two hosts and re-run: total traffic is about six HTTP requests per case-year, because sites are deduplicated to the weather grid before anything is asked for. Or hand-download a dashboard export into `data/raw/eirgrid/` and the Open-Meteo JSON into `data/raw/weather/`, and everything after the fetch runs unchanged.

One thing that needs solving either way. 82 of WP2033's generator records, carrying 7.8 GW, have no coordinate, and almost all of that is offshore: Dublin Array, Arklow, the three Codling sites, Kish and Bray, Skerd Rocks, Oriel. The onshore substation is the wrong coordinate for them, because the weather at Codling Bank is not the weather at its landfall, and ERA5-Land does not cover sea in any case. Those sites need real coordinates and the coarser ERA5 grid, which the module already supports.

Code: `profiles.py` → `fetch()`, `load_cell()`, `farm_curve()`, `solar_capacity_factor()`, `validate_wind()`, `Unavailable`

8.2 The generator that stands in for it

`synthetic.py` produces a deterministic year from the case file alone. It exists for two reasons: as insurance against exactly the blocker above, and so that a study can build counterfactuals no historical year contains, such as a worse wind year or a higher-penetration scenario.

The single detail that decides whether it is worth anything is spatial correlation. Independent per-bus noise passes almost every test anybody would write: the profiles land in $[0, 1]$, the load factor comes out at 30%, each bus looks plausible on its own. And it is useless, because 400 independent sites average out. The central limit theorem flattens the fleet into a nearly constant series, and a network that is never asked to carry a surge is never constrained.

So the wind field is built as a Gaussian random field with an exponential correlation decay in distance, driven through an Ornstein-Uhlenbeck process in time with a 36-hour synoptic timescale, mapped onto a Weibull wind speed by a Gaussian copula, and put through the same power curve the ERA5 path uses:

$$\rho(d) = \exp\left(-\frac{d}{L}\right), \quad L = 400 \text{ km}$$

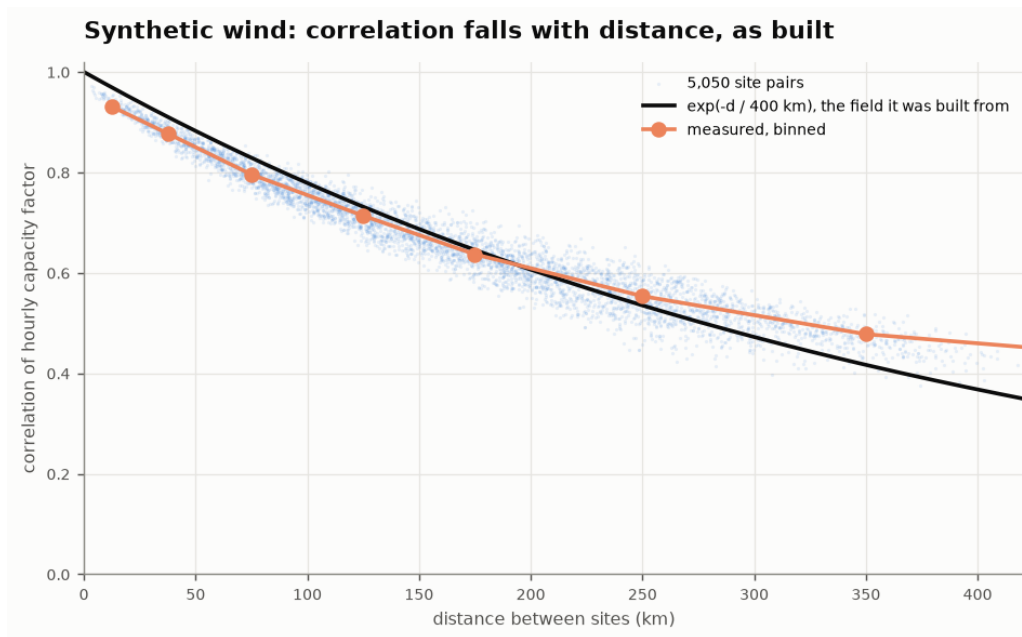


Figure 5: The check that matters, measured off the output rather than assumed. Every pair of wind sites, correlation of hourly capacity factor against distance. The black curve is the field it was built from and the orange line is what came out. They agree. Had this come out flat and near zero, every curtailment number in this report would be worthless.

The other half of the check is that the aggregate has to move. It does: fleet mean 0.313, standard deviation 0.248, 1,053 hours below 5% of capacity, 486 hours above 80%, and a longest continuous calm of 112 hours. Independent noise would give a standard deviation of a couple of points and would never approach either rail. Two tests assert both halves.

8.3 Anchored on the four published states

The generated year is not free-floating. WP2024 and WP2033 are winter-peak conditions and SV2024 and SV2033 are summer valleys, so each case is placed at an hour of the year that reproduces its state, and the field tapers back to unconstrained over 18 hours either side. Demand is rescaled so that the year's maximum and minimum are that vintage's own two totals, which keeps 2033's demand out of 2024's year.

```
ANCHOR_POINTS = {"WP": (1, 17, 18), "SV": (7, 9, 5)} # month, day, hour
ANCHOR_WINDOW_H = 18
SEED = 42
```

One anchor is deliberately refused. WP2033 dispatches 373 MW of solar at a winter-peak evening, a capacity factor of 0.10. At 18:00 on 17 January the sun is well below the horizon at every Irish latitude, so that value is unreachable and forcing it would put daylight into the profile at night. The generator declines it and records why in its anchor report. This is not a fault in the study files: a case is a set of conditions the network must withstand, with each carrier's output chosen for the study rather than for a clock.

The same reasoning cuts the other way and should be said plainly. The wind anchor is applied, so 17 January evening in the generated year is calm because the case says so. That is a security assumption being written into one hour of the weather, which is why the window is confined to 18 hours either side. A year in which every cold evening was calm would be a worse lie than the one this module replaces, and Irish winter peaks frequently coincide with high wind.

8.4 What is real in the generated year, and what is not

real	The network. Which generators exist, what they are, where they connect and what they are rated at. Which buses carry load and what share of the island's demand each one carries. Site coordinates. The four anchor states. The power curve and the panel model, shared with the ERA5 path.
generated	Every hour of weather. The shape of demand through the day. The hydro and thermal availability shapes. Where in the year the anchors sit.
solved rather than chosen	The Weibull scale, found by bisection so that the fleet load factor comes out at the target, which gives 8.40 m/s for the default settings. The demand scaling, so that the year's extremes are the case's own.

Table 10: The network, the fleet and where the demand sits are real. When anything happens is invented. A curtailment total computed from this is a statement about a plausible year rather than about any year that occurred.

8.5 Does the 2033 case actually bind?

The whole exercise turns on it. If the profiles do not produce hours where the network cannot take what the wind offers, the central problem never appears.

	WP2024	WP2033
hours studied (windiest)	60	60
energy offered	1.4 GWh	342.8 GWh
energy curtailed	0.0 GWh	20.4 GWh
curtailment	0.0%	5.94%
hours with curtailment	0 of 60	60 of 60

Table 11: It binds, in every hour studied. The 2024 comparison is not a fair fight and should not be read as one: the WP2024 network is built from in-service machines and that case runs almost no wind, so it holds a single wind generator and there are only 1.4 GWh to curtail. What the pair shows is that the scenario creates the problem and the profile generator does not.

The single binding circuit, at 100% of its rating in every hour, is 3581-89516-1, Letterkenny to Strabane. Section 7 reached the same circuit from a completely different direction, and Section 10.1 is about how much that agreement is worth.

Code: `synthetic.py` → `correlation_matrix()`, `gaussian_field()`, `anchors()`, `taper()`, `binding()`

9 Part VII. The participant kit

participant-kit/ is a standalone directory. It contains no build tooling and imports nothing from the rest of the repository, so it can be lifted out whole.

networks/	4 scenarios $\times$ 2 scopes, as netCDF and as CSV folders
gridkit.py	load, edit, reset, save, and read results back
flowmath.py	PTDF, susceptibility, shift factors
plotstyle.py	one validated palette for every chart
examples/	six scripts, a to f
test_kit.py	20 checks, runnable without pytest

scenario	buses	circuits	generators	week peak demand	capacity
WP2024	646	645	295	7,325 MW	22,650 MW
SV2024	639	640	225	4,948 MW	14,591 MW
WP2033	754	755	929	8,792 MW	42,574 MW
SV2033	754	755	264	6,058 MW	18,283 MW

Table 12: The all-island networks, read from `networks/manifest.csv`. The generator count includes one load-shedding generator per load bus, which is why it exceeds the machine count. Capacity excludes those. The four differ in fleet size because each network is built from its own case’s in-service records: WP2024 carries one wind generator and WP2033 carries 425. WP2033 is the interesting case, with 42.6 GW of connected plant against an 8.8 GW peak, and that headroom is where constraint and curtailment live.

Every network carries 168 hourly snapshots, one week centred on the hour that matches its own case state. Both file formats ship because many participants would rather edit a spreadsheet than learn an API.

9.1 The loader

```
import gridkit

gridkit.quiet()
n = gridkit.load("WP2033", "all-island")

gridkit.add_line(n, "Meentycat", "Srananagh 220", s_nom=200)
gridkit.remove_line(n, "3581-89516-1")
gridkit.set_rating(n, "2781-4951-1", 200.0)
gridkit.add_battery(n, "Letterkenny", p_nom=50, hours=4)

gridkit.solve(n)
print(gridkit.curtailment(n).head())
print(gridkit.binding(n).head())

n = gridkit.reset()
```

Code: participant-kit/gridkit.py $\rightarrow$ `load()`, `add_line()`, `remove_line()`, `set_rating()`, `add_battery()`, `reset()`, `curtailment()`, `binding()`, `unserved()`

9.2 Two traps, both already sprung

Both are in the kit’s README, because each cost time to find and neither announces itself.

Freezing the dispatch. `n.optimize()` writes its answer to `generators_t.p`. `n.lpf()` reads `generators_t.p_set`, which is a separate table that the kit ships empty. Run the power

flow straight after the optimisation and you get a flow with no generation in it: the reference bus silently supplies the whole island and one circuit shows at 1600% of its rating. `gridkit.freeze_dispatch(n)` copies one into the other.

Buses at 0°N 0°E. PyPSA's netCDF writer turns a missing coordinate into 0.0 instead of a null, and that point is in the Gulf of Guinea. Plot the raw bus table and Ireland ends up in the corner of an Atlantic-to-equator map. `gridkit.placed_buses(n)` drops those and respects the `coordinate_source` column.

9.3 (a) DC power flow

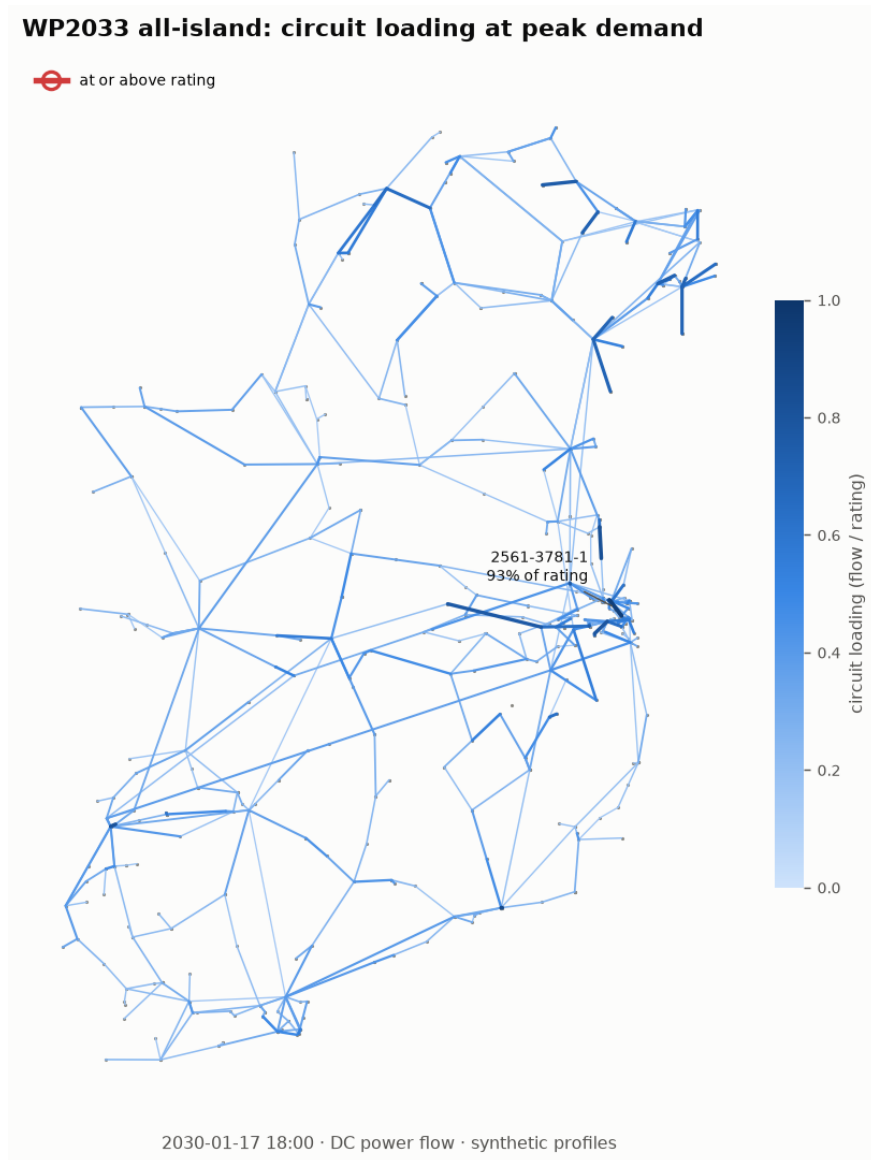


Figure 6: WP2033 at peak demand. Each circuit is coloured by how full it is, and the one at or over its rating is drawn in red and ringed, because several circuits join buses a few hundred metres apart and a stroke alone would be invisible. Nothing here is optimised: a DC power flow takes the dispatch as given and says where the flow ends up.

Six circuits sit at their rating for at least one hour of the week. The worst, 3581-89516-1, is at 100% for **118 of 168 hours**.

Code: `participant-kit/examples/a_dc_power_flow.py → main(), _draw()`

9.4 (b) Dispatch, curtailment, and how much of it the network caused

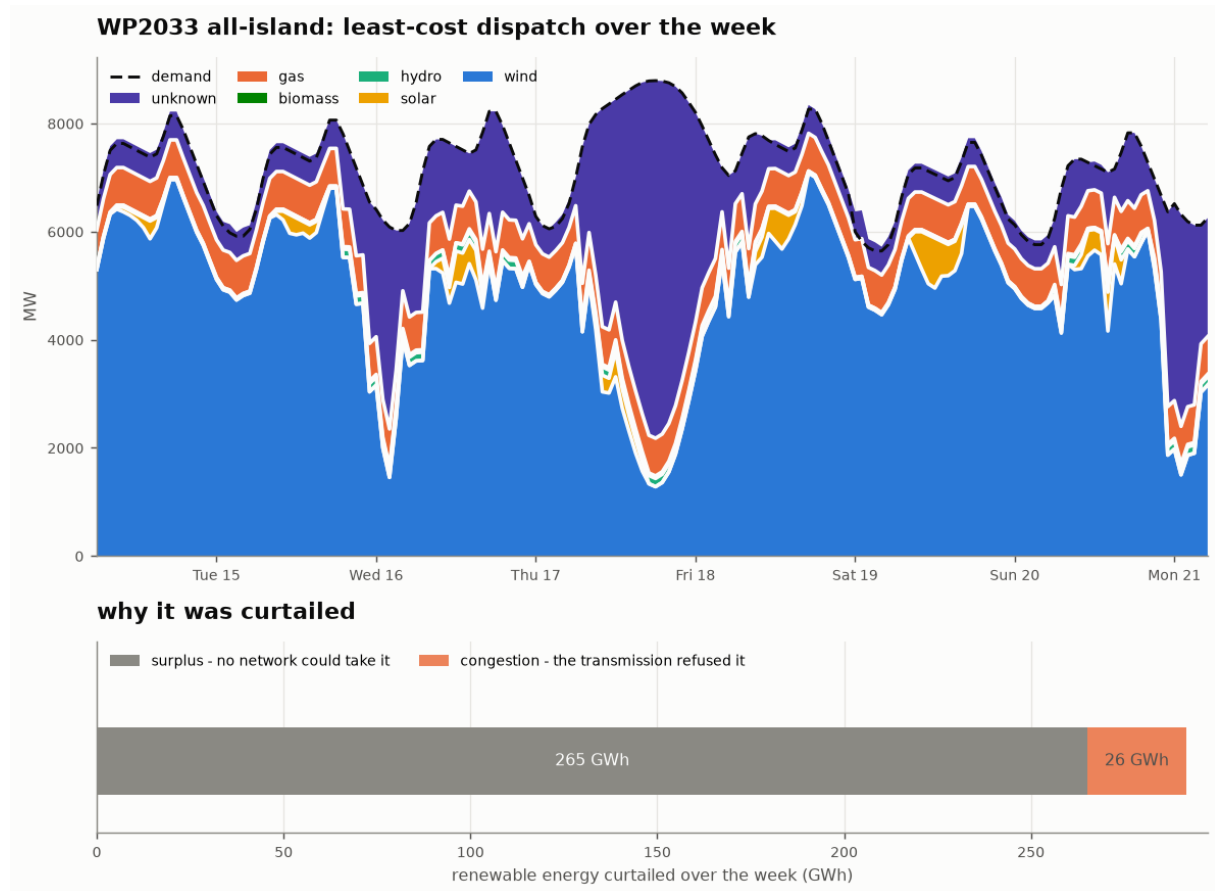


Figure 7: Least-cost dispatch over the week, and the reason for the curtailment.

Wind supplies 68% of the week's energy. 291 GWh of renewable output is curtailed, which reads as a damning verdict on the transmission system until it is separated.

The separation costs one extra solve. Lift every branch rating out of the way and re-optimize. Whatever is still curtailed is surplus that no network could have taken, and the difference is what the transmission cost.

265.2 GWh	91%	surplus, from 42.6 GW of plant against an 8.8 GW peak
26.3 GWh	9%	congestion, which is what the network refused

Table 13: A large curtailment figure does not by itself implicate the network. In WP2033, building a line recovers none of the 265 GWh. This distinction is the difference between a headline and a finding.

Code: `participant-kit/examples/b_lopf_dispatch.py` → `_congestion_share()`, `by_carrier()`

9.5 (c) Capacity expansion

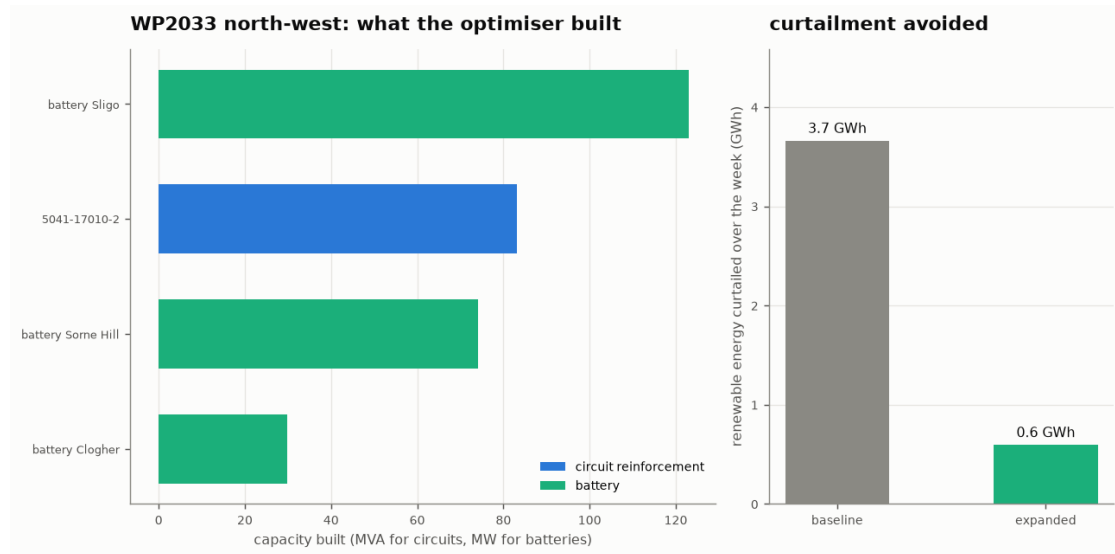


Figure 8: The binding circuits are marked extendable, a battery is offered at every renewable bus, both are given an annualised cost, and the optimiser trades building against curtailing.

On the North-West region it reinforces one circuit by 83 MVA and builds three batteries, cutting the week's curtailment from 3.7 GWh to 0.6. The costs are round numbers standing in for a price list. The point is the pattern: mark a component extendable, give it a capital cost, solve, and read the optimised rating back.

Code: `participant-kit/examples/c_capacity_expansion.py` → `_make_extendable()`,
`_offer_batteries()`

9.6 (d) PTDF from the pseudoinverse

Everything in `flowmath.py` comes out of one object, the weighted graph Laplacian $L = KBK^\top$, where K is the bus-branch incidence matrix and B the diagonal of branch susceptances.

L is singular, because adding a constant to every angle changes no flow, so it is inverted with the Moore-Penrose pseudoinverse:

$$\theta = L^+(p + KB\varphi), \quad F = B(K^\top\theta - \varphi), \quad \text{PTDF} = BK^\top L^+$$

Why the pseudoinverse rather than deleting a slack row. The textbook fix is to pick a slack, delete its row and column, and invert what is left. That works, and it writes the arbitrary choice of slack into every number that comes out. L^+ returns the solution orthogonal to the nullspace, which is the one where the balancing megawatt is spread evenly. The choice has not disappeared; it has become explicit, and `shift_factors()` lets you change it without rebuilding anything.

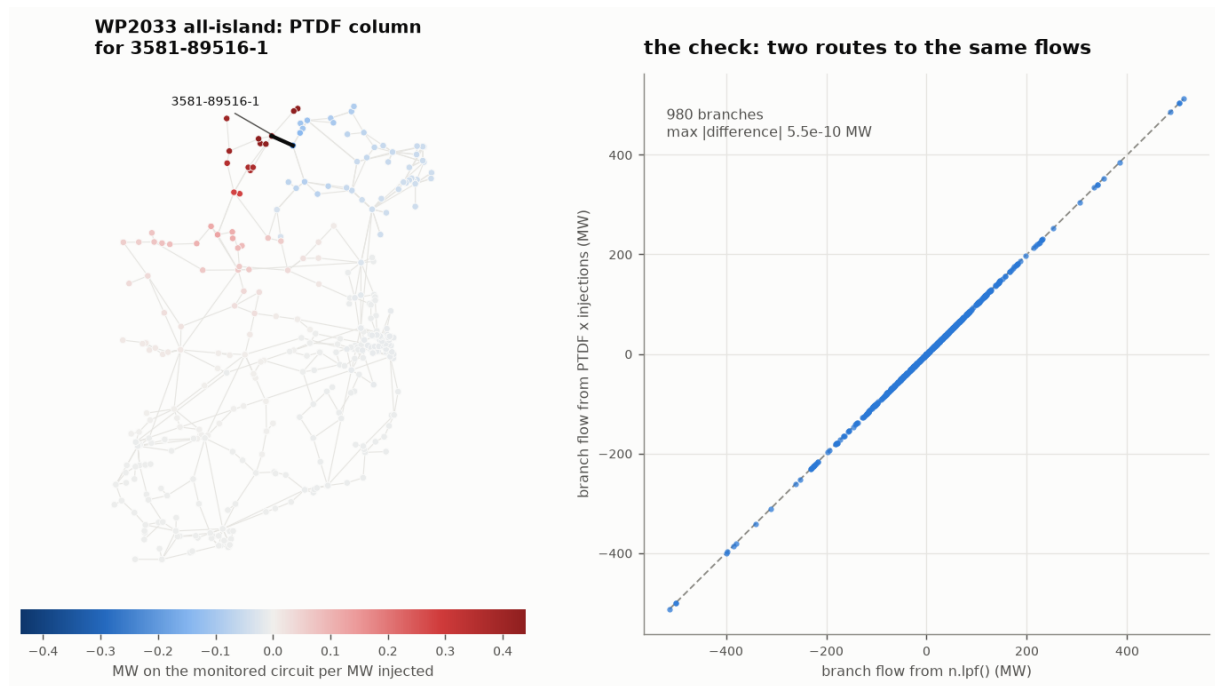


Figure 9: Left: one PTDF column on the map, showing how much of an injection at each bus the monitored circuit carries. The pale middle of the island is the part the circuit cannot feel at all. Right: the check. Branch flows rebuilt from $\text{PTDF} \times p$ against what PyPSA’s own linear power flow produced, over 980 branches. Maximum disagreement 5.5×10^{-10} MW.

That check earned its keep. It first came out at **79 MW**, and the reason was the network’s two phase-shifting transformers, one of them at 17° . A phase shifter imposes an angle of its own, so the flow is $b(\theta_i - \theta_j - \varphi)$ and the bus equation gains a $KB\varphi$ term, which behaves exactly like a pair of equal and opposite injections. The PTDF itself is unaffected and the flows are not. A PTDF that has not been checked against a power flow is a matrix of plausible numbers.

Code: `participant-kit/flowmath.py` → `branches()`, `laplacian()`, `pseudoinverse()`, `ptdf()`, `angles()`, `flows()`

9.7 (e) Edge-to-edge susceptibility, and Braess's paradox

Differentiating $F = BK^\top L^+ p$ with respect to one branch's susceptance gives the full edge-to-edge response matrix:

$$\frac{\partial F_e}{\partial B_{e'}} = \delta_{ee'} a_e - B_e \left(k_e^\top L^+ k_{e'} \right) a_{e'}$$

where $a_e = F_e/B_e$ is the effective angle difference across branch e , that is the plain angle difference less any phase shift.

The diagonal is the obvious part. The off-diagonal is where **Braess's paradox** lives [1]: a positive entry means that strengthening e' puts more power on e . The effect is established in power networks specifically, theoretically by Witthaut and Timme [22] and on a real AC laboratory grid with a topological predictor by Schäfer et al. [18]. Power in a network is imposed rather than routed. It divides between parallel paths in inverse proportion to reactance, and nothing consults the ratings, so building a second circuit anywhere in a mesh changes every impedance ratio in it.

WP2033 all-island at 2030-01-16 16:00: edge-to-edge susceptibility

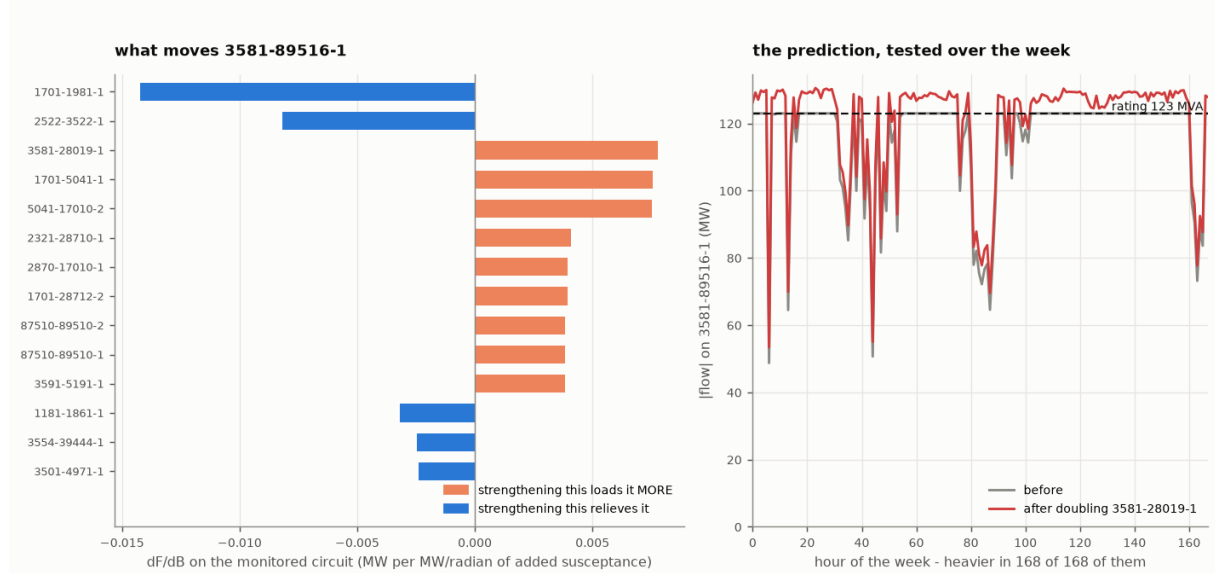


Figure 10: Left: the branches that move the island's most-loaded circuit, signed. Of the 979 other branches, strengthening 516 of them makes it worse. Right: the derivative is a claim, so the script tests it. It doubles the susceptance of the strongest candidate, which is what a second circuit on the same route does, holds the dispatch fixed, and re-solves the flow.

The result. Doubling 3581-28019-1 makes 3581-89516-1 carry more power in **168 of 168 hours**. Flow at the studied hour goes from 123.0 to 127.3 MW against a 123 MVA rating. A reinforcement a reasonable planner might propose makes the binding constraint worse.

Code: participant-kit/flowmath.py → susceptibility(), braess_candidates()

9.8 (f) Shift factors, and what a constraint group looks like

For one monitored circuit, how much of each generator’s output that circuit carries. Rank the generators by it and the result has the shape of a constraint group: the machines a dispatch instruction would have to act on.

Calling this a reproduction of EirGrid’s methodology would overstate it, and the heading above comes from the brief’s own wording. EirGrid’s published groups are defined by a monitored flow and are the product of contingency studies, an AC network and the Northern Ireland system [7]. What follows is a DC shift-factor ranking on one intact circuit. It is the right arithmetic and a smaller amount of it.

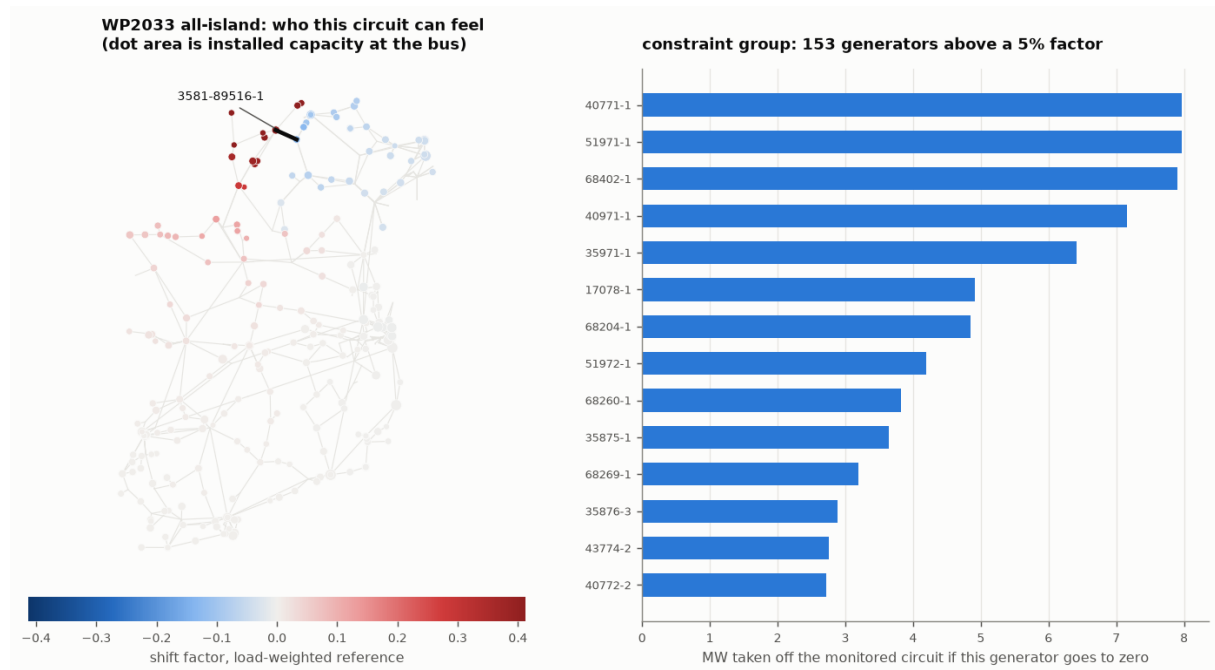


Figure 11: Left: the shift factor at every bus, with dot area showing installed capacity. Right: the constraint group, ranked by the megawatts of relief each member could actually deliver at the studied hour.

The thing to be careful about is the reference. A shift factor is megawatts on the circuit per megawatt at the generator, and a megawatt cannot appear on its own, so the definition is incomplete until you say where the balancing megawatt goes. The kit offers three conventions: spread it over every load in proportion to size, which is what a system operator means; spread it evenly over all buses, which is the pseudoinverse’s own reference; or put all of it at one named bus, which is the textbook single-slack convention.

The values differ and the ranking does not. Spearman rank correlation between the three references is **1.0000**. That invariance is why the method is robust enough to run a real dispatch tool on, and it is worth knowing that it is an invariance rather than a coincidence.

Code: `participant-kit/flowmath.py → shift_factors()`

10 Findings

10.1 Two independent methods agree on the binding circuit

This is the result worth the most, because nothing was tuned to produce it.

	Section 7	Sections 8 and 9
network	2024	2033
method	DC power flow, other ties pinned at the case's own values	full linear optimisation over a week
profiles	none, a capacity-factor sweep	generated correlated weather
scope	the North-West region	the whole island
rating of that circuit	93 MVA in WP2024	123 MVA in WP2033
answer	3581-89516-1, Letterkenny to the Strabane phase-shifter bus	

Table 14: Different network vintage, different method, different data, no shared code beyond the conversion itself. Same circuit.

How much this is worth, stated carefully. The two methods share the conversion, the geocoding and the topology. They are independent in vintage, dispatch, profiles and scope rather than in everything. The agreement is also partly structural: the region exports and this is its narrowest export path, so a thermal-limit method was always likely to land here. What the agreement rules out is that the finding is an artefact of the generated weather or of the 2033 build-out. It does not establish that this is the constraint a control room would act on.

Independent support from outside this work: EirGrid's two live North West constraint groups in February 2024 both monitor Letterkenny busbar flows [7], and the circuit found here is one of the circuits on that busbar. That corroborates the area rather than the specific circuit, and Section 7 explains why it cannot be more than that.

10.2 Most curtailment in the 2033 case is not the network's fault

Of 291 GWh curtailed in WP2033, 265 is surplus and 26 is congestion. The instinct on seeing a large curtailment figure is to propose reinforcement. In this scenario, 91% of it would be money spent on energy that has nowhere to go regardless.

10.3 Reinforcement can make a constraint worse, and here is a case

Of the 979 branches that could be strengthened, 516 would increase the flow on the island's most-loaded circuit. The strongest candidate, verified on a re-solved power flow, makes it worse in every hour of the week. This is a reason to compute the susceptibility matrix before proposing a circuit.

10.4 Below 38 kV, OpenStreetMap cannot supply a distribution network

Section 3. 2% of distribution transformers, 6,707 medium-voltage components against a true figure near 571, and no snapping tolerance that recovers the missing spans. Above 38 kV the same data is good and is free.

10.5 Four things about the hand-built North-West dataset

Restated because they are actionable: the hydro capacities are the 38 kV connection voltage rather than capacities; the 24-node and 15-node blocks differ by exactly Cunghill's 35 MW; splitting Srananagh back into two nodes takes the model's flow error from 21.6 MW to 1.7; and

the node set describes the 2024 network, coming apart in both 2033 cases where a new station replaces the Glenree to Moy circuit.

11 Bugs found, and what each one cost

Each of these produced a network that ran. That is the reason to write them down.

1. **Every optimisation infeasible.** PyPSA 1.x turns a non-null `p_set` into a hard equality that pins the dispatch. The networks carried the case's own dispatch, which is the case's answer asserted, and it is infeasible against a lossless model. Fixed by a function that clears it and keeps the case's values in a column of their own.
2. **The full network still infeasible.** The coupler-bounding rule, applied to sub-110 kV distribution transformers, produced limits that bind on the case's own flows on 13 transformers, worst at 156%. Fixed by restricting the rule to zero-impedance couplers.
3. **WP2033's angles out by 100°.** PyPSA's own slack choice survived a per-label assignment and one 21 MW hydro machine absorbed 856 MW. Section 5.6. Angle correlation 0.90 to 0.99.
4. **The optimisation exported 1,223 MW for profit.** A single bidirectional generator with a positive marginal cost and a negative lower bound gets paid to run backwards. Split into an import generator and an export sink.
5. **7.8 GW of wind running flat out.** 82 wind and solar generators in WP2033 had no availability series, because the generated weather is produced at geocoded sites and their buses never matched OpenStreetMap. PyPSA defaults a missing series to 1.0, so at a zero bid they ran at full output for all 168 hours: more than the island's peak demand, a fifth of the fleet, must-run baseload labelled wind. The curtailment figure read 64% instead of 26%. Each now borrows the nearest profile of its own carrier. **The dispatch chart looked entirely reasonable throughout.**
6. **The North-West had no wind at all.** The extraction folded every machine at a station into one generator with a single carrier, which was right for the reconciliation it was written for. For the kit it meant a region that is 90% wind could not curtail. Now split by carrier from the case's own machine records.
7. **Flows out by 79 MW.** The two phase-shifting transformers, caught only because the PTDF was checked against a power flow.
8. **A circuit at 1600% of its rating.** A power flow run after an optimisation without freezing the dispatch. Now a named function, with the symptom described in its docstring so the next person recognises it.
9. **Ireland in the corner of a map of the Atlantic.** Three buses at 0°N 0°E.
10. **Guards that never fired.** Several checks were testing the row count of an empty-column DataFrame, which is the snapshot count and always truthy, so the dispatch-freezing helper would silently freeze nothing.

12 Tests

```
python -m pytest                                # 215 passed
cd participant-kit && python test_kit.py          # 20/20, no pytest needed
```

file	tests	what it guards
test_network.py	37	the OpenStreetMap layer and the graph builder
test_profiles.py	32	the ERA5 pipeline, and that it refuses to invent
test_synthetic.py	31	correlation, anchoring, determinism
test_psse.py	30	the raw-file parser, against the files themselves
test_pypsa_net.py	26	every conversion decision in Section 5
test_northwest.py	21	the station map, the circuit count, both views
participant-kit/test_kit.py	20	the kit: networks, loader, linear algebra
test_geocode.py	18	normalisation, contraction codes, accept and reject rules
215		

Table 15: Counts from the last recorded full run. The machine this draft was written on has neither PyPSA nor pytest installed, so the suite was not re-run for it.

12.1 The two that matter most

Most tests check that code does what it says. These two check it against something that does not share code with it.

`test_ptdf_matches_a_power_flow` rebuilds every branch flow from $\text{PTDF} \times p$ and compares it against PyPSA’s own linear power flow. Two routes to the same number, agreeing to 10^{-6} MW. This is the test that found the phase shifters.

`test_susceptibility_matches_finite_differences` perturbs one branch’s susceptance by a ten-thousandth and checks the analytic derivative against the difference it actually makes:

```
numeric = (flows_with(bumped) - reference) / step
predicted = analytic.iloc[:, column].to_numpy()
# The natural scale is MW per MW/radian. Some columns are exactly zero,
# because a radial stub's susceptance changes no flow anywhere, so the
# column's own magnitude is not a usable denominator on its own.
scale = max(np.abs(predicted).max(), np.abs(reference).max() / base[column])
assert np.abs(numeric - predicted).max() / scale < 1e-4
```

If either drifts, nothing built on `flowmath.py` means anything.

13 Limitations

Read this before presenting anything from it.

This is a DC approximation of an AC model. The study files are full AC load-flow cases with reactive power, voltage magnitudes, tap changers and shunt compensation. All of that is discarded here: flows are linear in angle differences, voltages are 1.0 per unit everywhere, losses are zero. That is the standard approximation for transmission constraint work and it is good to a few percent on real flows, but it says nothing about voltage, reactive support or stability, and a conclusion that depends on those has no support here.

Ratings are the file's RATE1 column, and the reading of that column is an inference. See Section 5.2. Beyond that, a control room works to seasonal, weather-dependent and often dynamic limits that are not in these files, and runs circuits above the continuous rating for defined periods. A figure of 100% of rating here means 100% of a planning number under an inferred interpretation of which column that is. Ratings also change between scenarios: the circuit this report follows is rated 80, 93, 105 and 123 MVA across the four cases.

Line lengths are usable, with seven named gaps. 83% of transmission branches carry a length, it correlates at $r = 0.983$ with the great-circle distance between independently geocoded endpoints, and reactance per km separates cleanly into overhead line and cable. Seven real circuits in WP2024 carry no length, six of them in Northern Ireland, and a per-km calculation should skip those instead of treating a zero as a distance.

Every time series is generated. No hour in them ever happened. They are anchored so that each network's own scenario hour reproduces the case state, and the spatial correlation is measured off the output, but they are neither a historical year nor a forecast. Generators without a geocoded site borrow a neighbour's profile, which makes those pairs perfectly correlated when they should not be.

Costs are placeholders, and renewables bid negative. Gas 90, biomass 40, hydro 1 and imports 150 EUR/MWh, with wind and solar at -1 . The load-shedding generators bid 10,000 EUR/MWh, a round number chosen to sit far above everything else. It carries no relation to the Single Electricity Market's Value of Lost Load, which was EUR 12,533/MWh for 2022 [19]. The negative bid is deliberate, because it is how a support scheme makes a wind farm willing to pay to stay on, and it makes curtailment a last resort rather than a free choice. It also makes the objective negative, so only differences between two objectives mean anything.

About a fifth of generation carries the carrier unknown. Carriers are inferred from bus-name prefixes and a keyword list, because the files do not label plant by technology. Any result turning on the split between gas and unknown is unsupported.

The DC links are simplified. Fixed efficiency, no ramp limits, no minimum stable export, no market coupling. Interconnector flow here is what the cost assumptions make of it rather than what the day-ahead market would do.

One circuit binds, the far side is an infinite sink, and there is an unmodelled control device on the constrained path. The model prices export to Northern Ireland at nothing beyond the tie's thermal limit. In reality SONI, the Northern Ireland transmission system operator [20], would redispatch around it, and the tie terminates behind a phase-shifting transformer whose tap this model holds fixed (Section 7.6). What the result establishes is that the profiles produce binding constraints and where, not that any particular curtailment percentage is right.

The OpenStreetMap layers are a claim about mapping. Blank areas below 38 kV mean unmapped rather than unserved, and the county-to-county variation in mapped density measures mapper effort. Section 3 is the detail.

The study file data is EirGrid's, provided for reference purposes only. It describes a planning forecast. It does not describe the operational system, and it is published without warranty as to accuracy or fitness for any particular purpose. Nothing here is an EirGrid product or an EirGrid position.

14 What this draft changed, and what could not be checked here

This draft rewrote the previous one to cover the whole repository instead of the model build alone, and to put the provenance of every input in one place near the front. Section 2 is the part that did not exist before.

Numbers were re-derived where the environment allowed it. The machine this was written on has no scientific Python stack, so anything requiring PyPSA or pandas was carried across from the recorded runs and is attributed as such. What was re-derived directly from the source files, with a small standalone parser, is listed here so a reader knows which is which:

- The system base of 100 MVA and 50 Hz, from each case’s header line.
- The branch record counts and the equality of RATE1 and RATE2 across all 4,841 records, and the three values RATE3/RATE1 takes.
- The transmission branch counts, the length statistics, and the split between zero-length couplers and real circuits missing a length.
- The 18 unrated transmission branches in WP2024.
- The rating of the Letterkenny to Strabane circuit in all four cases.
- The transformer record assembly, checked by arithmetic, and the two phase-shifting transformers with both ends at transmission voltage.
- The geocoding method counts, placement rates and cross-check distances, from the committed CSVs.
- The OpenStreetMap candidate counts and the county sweep totals, from the committed CSVs.

Two corrections. The phase-shift angle at Strabane is 17° in the two winter cases and 12° in the two summer cases; the previous draft quoted 17° for all four. And the participant kit’s own README still carries the sentence “Line lengths are placeholders”, which Section 6.5 showed to be wrong. That file should be corrected to match.

14.1 What could not be verified from here

Stated plainly, because a citation the author could not open is weaker than one they could.

- **Every EirGrid, SONI and SEM-O host is refused by this environment’s egress policy**, as are the reanalysis and dashboard hosts. So the study file landing page, both Wind Dispatch Tool overviews and the Value of Lost Load paper are cited as located publications; their contents reached this document through a search index rather than through the documents. The group names in Section 7 are the claim most exposed to that.
- **The PSS/E format itself.** Siemens’ manual is not public. The bus type codes are corroborated by an independent open-source reader [17]; the rating semantics are not corroborated by anything, which is why they are presented as an inference.
- **Nothing in the study files was checked against the physical system.** Where this report says a circuit is rated at 93 MVA, it means the file says so.
- **ESB Networks’ asset counts** are quoted from its published description of its own network and were not independently checked.

15 Where everything is

file	lines	what it does
<i>the coverage study</i>		
network.py	766	the OpenStreetMap layer and the graph builder
eirgrid.py	379	EirGrid's asset register and county boundaries
analysis.py	541	the five measurements behind FINDINGS.md
plots.py	430	the national map and one per study area
extract_web_data.py	166	lines and point assets for the web map
extract_web_base.py	128	coastline and county outlines
build_web_map.py	377	one self-contained HTML file
<i>the model build</i>		
psse.py	675	reads PSS/E v35 raw files, Section 4
pypsa_net.py	1352	the conversion, Section 5
geocode.py	1085	OpenStreetMap matching, Section 6
northwest.py	958	the region, Section 7
profiles.py	948	the ERA5 pipeline, never run
synthetic.py	964	the generated year, Section 8
build_kit.py	459	assembles the participant kit
<i>the kit, which imports none of the above</i>		
participant-kit/gridkit.py	345	the loader
participant-kit/flowmath.py	340	PTDF, susceptibility, shift factors
participant-kit/plotstyle.py	99	the palette
participant-kit/test_kit.py	300	the checks
participant-kit/examples/		six scripts, a to f
<i>documents</i>		
FINDINGS.md	432	the coverage verdict, in full
docs/PHASE1_PARSER.md	292	the file format and its validation
docs/PHASE2_PYPSA.md	605	every conversion decision, with its evidence
docs/PHASE3_NORTHWEST.md	474	the reconciliation, table by table
docs/PHASE4_PROFILES.md	338	the ERA5 pipeline and the blocker
docs/SYNTHETIC_DATA.md	333	what is real and what is fabricated
docs/report/		this document, its figures and its sources

Table 16: This report is the summary. The phase documents are the working notes, and where a number here has a longer story, that is where it is.

15.1 Outputs worth knowing about

data/national_network.png	both layers over the Republic
data/ireland_distribution_map.html	the same distribution layer, interactive, opens from disk
data/eirgrid_transmission.gpkg	EirGrid's 110 kV-and-above lines and stations
data/analysis.json, national.json, county_sweep.csv	the coverage measurements
data/osm_substations.csv	the 1,705 named OpenStreetMap objects
data/pypsa/<case>_<scope>/	the matching runs against each case as a PyPSA CSV folder, with a report table per decision
data/pypsa/geocoding/<case>.csv	where each bus is, how it was matched, and why the rest were not
data/pypsa/northwest_<case>_<view>/	the region in both views, with circuits, boundary flows and the comparison against the full network
participant-kit/networks/	the eight shipped networks

Reproducing every figure in this document.

```
python plots.py national          # the map in Part I
python docs/report_figures.py     # the four build-half figures
cd participant-kit
for e in examples/*.py; do python "$e"; done # the six kit figures
cd ../docs/report && ./build.sh      # pdflatex, bibtex, pdflatex x2
```

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